
From Signal Fusion to Asset Allocation: A Decision-Theoretic Model for Portfolio Construction Under Regime-Based Sentiment and Volatility

Author 1

Department withheld for blind review,
Institution withheld for blind review,
City withheld for blind review
E-mail: withheld for blind review

Author 2

Department withheld for blind review,
Institution withheld for blind review,
City withheld for blind review
E-mail: withheld for blind review

Abstract: This paper deals with a critical but relatively under-addressed question in the field of quantitative finance: how to move from the activity of generating predictive financial signals to the activity of executing decisions in asset allocation. The current literature dwells mainly on predictive models based on technical indicators and sentiment-based measures, and does not consider the systematic construction of methodologies that convert those models into effective portfolio weights in dynamic market regimes. A tripartite architecture is created through the proposed methodology. First, a set of multimodal signals, including those derived from technical, sentiment and volatility information, is synthesised. Thereafter, the latent market regimes are outlined using Gaussian mixture models, which in effect define different probabilistic conditions of the financial environment. The central mechanism is a regime-conditional fusion rule that calibrates the impact of each signal by its assumed effectiveness within the identified state, a hypothesis this study tests rather than presumes. This culminates in a decision-theoretic core of optimisation, which incorporates shrinkage covariance structures and regime-dependent risk-budgeting parameters in converting the synthesised signal into a portfolio construction to be implemented. Evaluated on ten liquid multi-asset instruments over eighteen years under a walk-forward protocol with transaction costs, the strategy attains a Sharpe ratio of 0.88 against 0.63 for an equal-weighted benchmark and lowers maximum drawdown from -36.1 to -19.0 per cent, at the cost of a lower compound return. The advantage holds in all eighteen specifications examined, in all four stress episodes, and in a held-out final block of 665 sessions that postdates every specification decision, but it is not statistically significant. Ablation at matched exposure locates the source: timing risk by market state contributes, whereas the signal fusion layer does not. Considering this, this study is relevant to applied decision sciences because it provides a clear and systematic way of closing the gap between predictive signals and actionable investment decisions, together with evidence on which parts of such a framework carry the result.

Keywords: Portfolio Optimisation; Regime Detection; Sentiment Analysis; Risk Budgeting; Decision-Theoretic Framework; Signal Fusion; Ablation Analysis; Asset Allocation

Reference to this paper should be made as follows: Author 1 and Author 2 (20XX) ‘From Signal Fusion to Asset Allocation: A Decision-Theoretic Model for Portfolio Construction Under Regime-Based Sentiment and Volatility’, *Int. J. Applied Decision Sciences*, Vol. X, No. X, pp.xxx–xxx.

Biographical notes: Biographical notes for the first author are withheld to preserve anonymity during review. The author’s research interests include computational finance, machine learning for financial time series, and decision-theoretic portfolio construction. The author conceived and implemented the framework reported in this paper.

Biographical notes for the second author are withheld to preserve anonymity during review. The author’s research interests include machine learning, data analytics and their application to financial decision problems. The author supervised and validated the study reported in this paper.

1 Introduction

The proliferation of computational finance methodologies has fundamentally transformed investment decision-making processes, with machine learning and artificial intelligence emerging as dominant paradigms for financial market analysis. Contemporary research has demonstrated considerable sophistication in predictive modelling, particularly through the integration of diverse data modalities including technical indicators, market microstructure data, and alternative data sources such as social media sentiment and news analytics (Li and Bastos, 2020; Zhang et al., 2022). Despite these advancements, a critical disconnect persists between the generation of predictive signals and their systematic translation into executable portfolio allocation decisions—a gap that remains largely unaddressed in the current literature.

1.1 Current State of Predictive Modeling in Finance

The evolution of financial forecasting methodology has been a continuing movement towards greater sophistication and the systematic combination of heterogeneous data modalities. Initial models were largely based on technical analysis, seeking to predict future price movement using historically derived price indicators and geometric pattern recognition (Lee et al., 2007). The subsequent arrival of machine learning pushed the field through a complete change. The introduction of algorithms such as support vector machines, ensemble techniques and multilayer perceptrons, in order to increase the accuracy of market predictions, marked that change (Chou and Nguyen, 2018; Liu and Wang, 2019). Deep-learning systems now constitute the modern frontier and have shown a strong ability to capture the complex nonlinear relationships that exist within high-frequency financial data (Nayak et al., 2022).

Another equally significant trend has been the methodical integration of sentiment analysis into predictive models. Early attempts in this direction used simple dictionary-based models (Loughran and McDonald, 2011), but more recent approaches progressively use transformer-based large language models (LLMs) to extract complex sentiment and thematic indications from unstructured financial text (Jung and Jang, 2024; Liu et al., 2024). Empirical studies such as those of Mu et al. (2023) and Bacco et al. (2024) show that quantitatively well-founded measurements of investor sentiment can considerably increase the explanatory value of traditional technically based models. Going beyond sentiment, Ónozó et al. (2024) point out the emerging capacity of LLMs to support the nowcasting of macroeconomic features, which extends the range of predictive characteristics that can be quantitatively analysed.

1.2 The Signal Fusion Paradigm and Its Limitations

A current trend in modern quantitative finance research is to combine heterogeneous types of signals in order to arrive at a stronger forecasting model. An example is the study of Moodi et al. (2024), combining technical indicators and sentiment analysis into a hybrid deep-learning model. In a similar manner, Anbaee Farimani et al. (2024) designed a multimodal adaptive learning framework that dynamically synthesises heterogeneous sources of data to predict financial markets. Such fused methodologies are consistently shown to perform better in prediction than their uni-modal counterparts, and are particularly effective in describing the multifaceted, multi-dimensional drivers of market dynamics.

Despite these substantive methodological developments, an overwhelming deficiency stands out in the body of extant research: a near-exclusive focus on predictive accuracy, at the cost of the decision-making mechanisms that are necessary to put a forecast into practice. With statistical forecasting precision as the primary measure of success, the predominant notion of success—as Zhang et al. (2022) note in their systematic review of decision fusion—is not actual economic value or achieved portfolio performance. The critical discontinuity that arises out of this singular interest in prediction is what we refer to herein as the *signal-allocation gap*: the gap between the production of informative signals and their encoding into a rational process of portfolio construction that takes account of real-world investment restrictions and investment goals.

1.3 The Regime Dependency Challenge

Financial markets are dynamic in nature, evolving over discrete structural regimes, whether of high volatility, persistent trend or mean reversion, that inherently condition the behaviour of trading signals and investment strategies (Wang et al., 2019). An important gap in the modern literature is the overwhelming oversight of these regime requirements. Even though the presence of such market states is frequently accepted, its systematic integration into portfolio-forming structures remains significantly underdeveloped (Kritzman et al., 2012).

A number of studies have attempted to deal with market-state variability. As an example, Huang et al. (2020) studied dependencies using bicluster mining and fuzzy inference to arrive at automated trading predictions. Similarly, Alsheebah and Al-Fuhaidi (2024) modelled various market conditions in an emerging economy by combining endogenous and exogenous variables. More recent work has taken the state variable directly into the allocation decision (Li et al., 2025; Luo and Mulvey, 2026). The methodologies of the predictive strand, however, are typically limited to increasing predictive accuracy and do not advance to regime-aware portfolio optimisation.

Accordingly, the key issue is twofold: it involves not only the determination of the prevailing market regime, but also the dynamic re-tuning of both the signal-fusion parameters and the underlying optimisation dynamics in response to the properties of the diagnosed regime.

1.4 Portfolio Construction: The Neglected Frontier

Modern studies show a significant lack of methodologies focused on the derivation of portfolio allocations from predictive signals. An intelligent method of constructing portfolios based on the prediction of financial returns, named IntPort, was proposed by Sami et al. (2025); the approach is, however, based on statistical averaging heuristics, which is insufficiently analytical to constitute a modern optimisation model. In line with this, most reinforcement-learning applications in automated trading, such as the one Kabbani and Duman (2022) describe, focus on the trading of individual assets rather than on the allocation of a portfolio.

This lack of explicit research on portfolio construction is particularly worrying given the underlying principles of modern portfolio theory, which emphasise the imperative of the covariance structure and the diversification of risk (Markowitz, 1952). The existing failure to convert individual asset predictions systematically into a single allocation method is a significant obstacle to the improvement of risk-adjusted returns. Moreover, the traditional methodology largely ignores critical practical restrictions, including transaction costs, position size limits and liquidity provision, which are inseparable from the implementation of a strategy in live investment settings (DeMiguel et al., 2009).

1.5 Our Contributions

This study bridges the identified methodological gaps through several pivotal contributions to financial decision science:

- An explicit regime detection framework employs Gaussian mixture models to isolate discrete market states from volatility and momentum data, extending the principles of Wang et al. (2019). The implementation integrates this regime intelligence into the subsequent signal fusion and optimisation stages, and the fitted model is reported with order selection, transition matrix and run lengths rather than asserted.
- A decision-theoretic fusion mechanism recalibrates the influence of technical and sentiment indicators according to their conditional performance within each identified regime, in place of a static weighting scheme. The paper specifies this mechanism in full and then tests it: the state-dependent weighting hypothesis is *not* supported by the evidence, and we report that result rather than the specification that flatters it.
- The research formulates a regime-contingent optimisation paradigm. This framework uses a shrinkage covariance estimate (Ledoit and Wolf, 2004), an explicit turnover penalty and a regime-dependent risk budget, progressing beyond the static assumptions of conventional mean-variance analysis. The fused signal is converted to a return metric before it enters the objective, so that the risk model and the signal are commensurate.
- A backtesting protocol with walk-forward validation is deployed, featuring regime-aware performance attribution to delineate the contribution of each strategic component under varying market environments. Components are separated by ablation at matched

average exposure, which isolates the effect of timing risk from the effect of simply holding less.

- An out-of-sample holdout is reported separately. The final 665 sessions of the sample postdate every specification decision taken in this study, and are evaluated only once, which addresses the concern that a walk-forward backtest reported in full can still be tuned in aggregate (Harvey and Liu, 2015; López de Prado, 2016).
- Ultimately, the proposed methodology constructs a transparent, systematic pipeline to traverse the critical divide between multi-modal signal prediction and executable allocation, directly countering the signal-allocation gap highlighted by Zhang et al. (2022).

Collectively, these contributions furnish both a methodological architecture and an empirical evaluation of a regime-conscious strategy. The framework improves risk-adjusted performance over an equal-weighted benchmark on every measure reported, though not significantly, and does so by timing risk rather than by selecting assets. Section 2 reviews the relevant literature, Section 3 specifies the framework, Section 4 reports the evidence and Section 5 concludes.

2 Literature Review

2.1 Evolution of Predictive Modeling in Financial Markets

Financial market forecasting has experienced a substantial development of methodology, which has progressed over the years from primitive technical analysis to highly developed multimodal learning architectures. The initial research was mainly based on technical indicators and statistical methods. As an example, Lee et al. (2007) established that multi-agent Q -learning systems could be used for daily equity trading, the limiting aspect at that time being computational capacity and the availability of data.

The introduction of deep learning initiated a paradigm change in the process of forecasting. Later works, including the numerical attention mechanism for dual-source information processing proposed by Liu and Wang (2019) and the sliding-window metaheuristic-optimised regression for price prediction proposed by Chou and Nguyen (2018), had significantly higher predictive accuracy. The major shortcoming of these methods, however, was that they viewed financial markets as stationary processes, thus largely ignoring the dynamic nature of markets in terms of regimes.

The latest level of research is marked by the creation of complex hybrid models. Nayak et al. (2022) demonstrated how an enhanced chemical-reaction-optimisation algorithm can be used in combination with dendritic neuron models, and therefore the potential of bio-inspired computing. At the same time, Anbaee Farimani et al. (2024) made a substantial contribution to the sphere, developing an adaptive multimodal model for the dynamic combination of various information sources, which is a significant step in the comprehensive management of the intrinsic heterogeneity of financial information. Related strands extend the same modelling apparatus to adjacent problems: forecast-horizon model selection for commodity prices (Zhang et al., 2020), empirical-mode decomposition for short-horizon series (Yang and Yang, 2020), penalised multinomial models for return direction (Hu and Yang, 2024), and heterogeneous-source adversarial learning for financial applications (Khuwaja et al., 2023).

This is described in Table 1, which shows the evolution from early computational models to present-day integrative models. Every new generation is more advanced in its capacity to process data, but one constant remains: the inability to connect predictive indicators to practical portfolio building.

Table 1 Evolution of Financial Prediction Methodologies

Era	Primary Methods	Key Contributions	Limitations
Early Computational (2004–2010)	Multi-agent systems, Q -learning, hybrid radial basis function (RBF) networks	Lee et al. (2007) – multi-agent Q -learning; Lee (2004) – iJADE Stock Advisor with hybrid RBF	Limited data sources, computational constraints, simplistic market assumptions
Deep Learning Revolution (2011–2019)	Long short-term memory (LSTM) and gated recurrent unit (GRU) networks, attention mechanisms, ensemble methods	Liu and Wang (2019) – numerical attention methods; Chou and Nguyen (2018) – metaheuristic optimisation	Stationary market assumptions, limited regime adaptation, focus on single-asset prediction
Multimodal Integration (2020–Present)	Transformer architectures, multimodal fusion, LLM integration	Moodi et al. (2024) – hybrid deep-learning fusion; Anbaee Farimani et al. (2024) – adaptive multimodal learning	Signal-allocation gap, insufficient portfolio integration, limited decision-theoretic framing

2.2 Sentiment Analysis and Alternative Data Integration

The use of sentiment analysis and other data streams constitutes one of the most significant changes in modern financial forecasting. Early methodologies mostly used lexicon-based methods and simple scoring procedures, of which the finance-specific word lists of Loughran and McDonald (2011) are the standard reference. Modern methods, though, increasingly exploit large language models and transformer networks to extract subtle sentiment, as in the case of Jung and Jang (2024), who applied targeted masking of numerical-change expressions and achieved significant improvements in the accuracy of sentiment quantification.

Liu et al. (2024) offer a comprehensive survey of large language models in financial market sentiment analysis, with critical analysis of the current approaches and datasets. Their synthesis highlights the transformative ability of such models to interpret financial discourse and, at the same time, illustrates the ongoing obstacles in terms of model interpretability and computational requirements. Similarly, Ónozó et al. (2024) demonstrate that the predictive capabilities of large language models can be useful in sentiment analysis and may also be used in the nowcasting of macroeconomic variables, which increases the range of predictive inputs available to quantitative modellers. Within this journal, George and Murugesan (2025) compare hybrid feature extraction across fourteen classifiers for equity sentiment, and report that the gains depend more on the feature representation than on the classifier.

The power of sentiment integration is still confirmed by empirical evidence. Mu et al. (2023) developed a stock-price prediction model which combines investor sentiment and optimised deep learning, and reported better performance than models based on traditional technical indicators alone. In further support of this result, Bacco et al. (2024) investigated stock prediction using LSTM networks and sentiment analysis of tweets in a context of

strong market uncertainty, which confirms the increased informational utility of sentiment in high-volatility environments. Parallel evidence comes from asset classes outside equities, including cryptocurrency price prediction (Parekh et al., 2022; Otabek and Choi, 2022), Bitcoin range forecasting from high-dimensional indicators and social dynamics (Shang, 2025), and electricity price prediction assisted by sentiment agents (Lu et al., 2025).

A distinct and considerably older tradition measures sentiment from market data rather than from text. Baker and Wurgler (2007) construct an investor sentiment index from observable market quantities—trading volume, volatility and issuance activity—on the argument that sentiment reveals itself in behaviour that can be priced, not only in language that must first be classified. The framework proposed in this paper belongs to this market-derived family, a point developed in Section 3.1.3 and validated externally in Section 4.3.3.

Table 2 provides an overview of the most common sentiment-analysis strategies used in financial forecasting, tracing the history from unsophisticated lexicon-based methods to solutions using large language models. Each successive category shows incremental improvement in handling the complexities of financial language, despite long-standing issues concerning computational efficiency and transparency of the model.

Table 2 Sentiment Analysis Methodologies in Financial Prediction

Methodology Category	Key Studies	Technical Approach	Reported Benefits	Identified Gaps
Lexicon-Based Approaches	Loughran and McDonald (2011)	Dictionary-based scoring, rule-based classification	Computational efficiency, interpretability	Limited context understanding, poor handling of financial jargon
Traditional machine learning (ML) methods	Vargas et al. (2022); Wang and Chen (2024)	Support vector machines (SVM), random forests, feature engineering	Improved accuracy over lexicons, better handling of context	Manual feature engineering, limited scalability
Deep Learning Architectures	Mu et al. (2023); Bacco et al. (2024); Tai et al. (2022)	LSTM, GRU and convolutional neural network (CNN) layers with embeddings	Automatic feature learning, context preservation	Computational intensity, black-box nature
LLM and Transformer-Based	Jung and Jang (2024); Liu et al. (2024); Ónozó et al. (2024)	Fine-tuned transformers, attention mechanisms, prompt engineering	Superior contextual understanding, multi-task capability	High computational cost, interpretability challenges
Market-Derived Measures	Baker and Wurgler (2007); this study	Volume, volatility and participation aggregates	No text pipeline, no language dependence, fully causal	Coarser than text; cannot attribute sentiment to a named event

2.3 Signal Fusion and Decision Integration Frameworks

The integration of various signal typologies forms a major model in advanced financial forecasting systems. The systematic review by Zhang et al. (2022) of decision fusion for equity prediction, synthesising evidence from 76 primary studies, shows that although fused-model architectures usually outperform unimodal ones, a high degree of diversity remains in both fusion methods and evaluation protocols.

A range of integrative architectures has been developed in recent research. Moodi et al. (2024) built a hybrid deep-learning system integrating technical indicators and sentiment analysis, demonstrating the complementary nature of different data representations. Further developing the field, Haryono et al. (2023) presented an approach using transformer-gated recurrent units to synthesise news sentiment with technical indicators, which emphasises the relevance of attention mechanisms in the fusion process, and Haryono et al. (2024) extended the same line with aspect-based sentiment generation. Systematic reviews of the deep-learning and technical-analysis literature report the same pattern of accumulating architectural sophistication (Li and Bastos, 2020; Wang et al., 2020).

Despite such methodological developments, a basic constraint persists, referred to here as the signal-allocation gap. This weakness is a significant point of discontinuity between sophisticated prediction models and their application to the creation of a portfolio. As Zhang et al. (2022) note, the most common measure of success is still the level of statistical forecasting accuracy, rather than economic usefulness or achieved portfolio return. The failure is acutely apparent in the reinforcement-learning case, such as that of Kabbani and Duman (2022), which largely considers single-asset trading and thus does not address allocation at the portfolio level.

A related literature concerns how such claims should be evaluated at all. Harvey and Liu (2015) show that the number of specifications a researcher may examine before reporting one inflates apparent performance to a degree that conventional significance thresholds do not accommodate, and López de Prado (2016) makes the parallel argument for portfolio construction specifically, that a weighting scheme selected on in-sample evidence frequently fails out of sample. Within this journal, Zhang et al. (2025) report directly that the model selected as optimal in sample is not reliably the model that performs best out of sample. These findings motivate two design decisions taken in Section 3: the full specification grid is reported rather than its best cell, and a block of sessions postdating every specification decision is held out and evaluated once.

2.4 *Market Regime Detection and Adaptation*

Recognition of market regime dependencies is an essential change in the field of financial modelling. The foundations of regime-conscious models were formed by Wang et al. (2019), who explored economic recession prediction on the basis of multiple behavioural factors. Their results verify that different market settings are linked with specific behavioural signatures, and hence require distinct methods of analysis. The econometric foundation for this view is older still: Hamilton (1989) formalised the notion that a single series may be generated by distinct, unobserved states with their own dynamics, and Kritzman et al. (2012) carried the idea into allocation by showing that a strategy conditioned on the inferred state can dominate its unconditional counterpart.

Regime considerations have been added at different levels in subsequent scholarship. Huang et al. (2020) proposed an automated trading-point prediction framework using bicluster mining and fuzzy inference, thereby embedding market-state dependencies implicitly in the form of pattern recognition. Similarly, Alsheebah and Al-Fuhaidi (2024) combined endogenous and exogenous variables to accommodate disparate conditions in emerging markets, though their main aim was predictive accuracy rather than asset allocation. More recent work places the state variable inside the allocation problem itself: Li et al. (2025) combine a jump model with model predictive control, and Luo and Mulvey (2026) condition both the signal and the investable set on a dual-regime indicator. A closely

related mechanism scales total risk by recent volatility without conditioning on a discrete state at all (Moreira and Muir, 2017; Hood and Raughtigan, 2025); this study separates the two so that their contributions can be compared, and Section 4.3.4 reports the comparison.

Table 3 provides a comparative analysis of the portfolio construction strategies reported in the literature, and displays how earlier methods based on conventional optimisation techniques have been succeeded by modern multimodal and risk-based approaches. However, in spite of the obvious rise in methodological sophistication, substantial gaps persist, especially concerning dynamic regime adaptation and the systematic transformation of predictive signals into the allocation choice. The present study sits between the two responses that the literature has developed to the instability of mean-variance inputs: robustification of the inputs themselves, as in the multi-interval robust mean-conditional-value-at-risk formulation of Khodamoradi et al. (2023), and abandonment of the expected-return term altogether in favour of a purely risk-based construction. It retains an expected-return term, but scales that term explicitly so that its magnitude is commensurate with the risk term, and it separates the composition decision from the total-risk decision so that the two can be evaluated independently.

Table 3 Portfolio Construction Methodologies in Literature

Methodology	Representative Studies	Basis of Allocation	Principal Limitation
Traditional Optimisation	Markowitz (1952)	Expected returns and covariance	Extreme sensitivity to estimated inputs
Naive Diversification	DeMiguel et al. (2009)	None; equal weights	Ignores risk structure entirely
Shrinkage-Based Optimisation	Ledoit and Wolf (2004)	Regularised covariance	Still requires expected returns
Signal-Based Allocation	Sami et al. (2025) – IntPort	Statistical averaging, forecast-based weighting	No covariance consideration, limited risk management
Risk-Based Construction	López de Prado (2016); Asimit et al. (2025)	Risk model only	Forgoes signal information by design
Volatility Management	Moreira and Muir (2017); Hood and Raughtigan (2025)	Total risk scaled by recent volatility	Does not address composition
Regime-Switching Allocation	Kritzman et al. (2012); Li et al. (2025); Luo and Mulvey (2026)	State-dependent parameters	Component contributions rarely isolated
Reinforcement and Learning-Based	Kabbani and Duman (2022); Ha et al. (2026)	Learned policy or predicted returns	Single-asset focus; interpretability
Multimodal Fusion Approaches	Moodí et al. (2024); Anbaee Farimani et al. (2024)	Multiple signal integration, advanced ML architectures	Prediction-allocation gap, no explicit regime adaptation
This study	—	Regime-conditional signal, plus a relative regime risk budget	Signal layer tested and found not to contribute

3 Proposed Methodology

The framework is organised in three phases, together with a set of cross-cutting properties that apply to all of them. Phase 1 extracts multi-modal signals from the raw data, Phase 2 fuses those signals conditionally on the identified market regime and converts the fused view into portfolio weights, and Phase 3 evaluates the result and attributes it to its components. The level of detail below is deliberate, so that the study can be reproduced independently from this description alone.

3.1 Phase 1: Multi-Modal Signal Extraction

3.1.1 Input Layer

The architectural framework initiates with the processing of market data for the target securities, structured as open-high-low-close-volume (OHLCV) series:

- **Market data.** Daily open, high, low, close and volume records for the instruments in the study universe, adjusted for splits and dividends, obtained from Yahoo Finance through the `yfinance` interface. The download window is 1 January 2007 to 27 August 2026. The end date is not a modelling choice: it is every session available at the date the analysis was run, so the sample boundary carries no information about the results.
- **News and social media.** Not used. No news article, social media post, corporate filing, transcript or analyst note is read at any point in this study. The reason, and the measure adopted in place of a textual one, are given in Section 3.1.3.
- **Macroeconomic data.** Not used as an input. Published macroeconomic and sentiment series are employed once, in Section 4.3.3, and there only as external validators that never enter the framework.

These raw inputs undergo a preprocessing pipeline involving cleansing, temporal normalisation and synchronisation to construct a cohesive, time-aligned dataset for subsequent analytical stages.

The primary study universe consists of ten exchange-traded funds selected to span the major liquid asset classes available to a diversified allocator: US large capitalisation equity (SPY), US large capitalisation growth equity (QQQ), US small capitalisation equity (IWM), developed international equity (EFA), emerging market equity (EEM), long-maturity US Treasuries (TLT), intermediate-maturity US Treasuries (IEF), investment-grade corporate credit (LQD), gold (GLD) and US real estate (VNQ). Breadth matters for the research question. A framework whose stated purpose is to redirect risk across market states can only be tested where there is somewhere else for risk to go; a single-sector universe offers no such destination, a point returned to in Section 4.3.6.

Indicator warm-up and the 252-session covariance estimation window consume the first year, so the evaluated out-of-sample period runs from January 2008 to August 2026 and covers the global financial crisis, the 2011 sovereign debt episode, the 2020 pandemic crash, the 2022 inflation-driven bear market and the subsequent recovery. Sessions from 1 January 2024 onward are treated as a holdout. That block postdates the design of the framework and every specification decision reported here; none of it was examined while the model was being built. It enters the full-sample results for completeness and is reported separately in Section 4.3.1. A secondary universe of ten mega-capitalisation US equities over 2018–2026 is carried through the same pipeline and reported in Section 4.3.6.

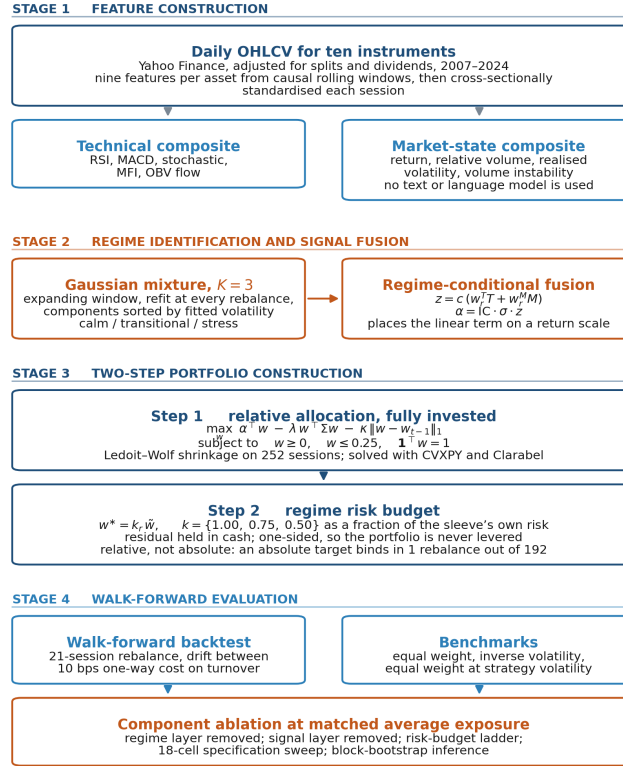


Figure 1 Proposed Methodology. Phase 1 constructs causal, cross-sectionally standardised features; Phase 2 identifies the market state, fuses the two composites and separates relative composition from total risk; Phase 3 evaluates the result and ablates its components.

3.1.2 Technical Indicators Engine

This component transforms raw market data into quantitative trading signals through three primary analytical categories. Nine features are computed for each asset on each session. All are causal: every value at session t uses only information available at the close of session t .

- **Price-derived metrics.** The relative strength index (RSI) identifies potential overbought or oversold asset conditions. The moving average convergence divergence (MACD) line captures trend momentum shifts. The stochastic oscillator locates the close within its recent trading range.
- **Volume-analysis tools.** On-balance volume (OBV) assesses momentum through cumulative volume flow, entered as a twenty-session change normalised by average volume, while the money flow index (MFI) combines price and volume into a single participation measure. Relative volume measures turnover against its own recent average.

- **Volatility estimators.** Realised volatility measures the statistical dispersion of asset returns over a specified look-back period, and volume instability measures the dispersion of daily volume changes.

No Bollinger band, average true range or volume profile construction is used; the nine features of Table 4 are the complete input set.

Working mechanism: a sliding-window methodology facilitates all calculations, with parameters calibrated for the horizons given in Table 4. No other input enters the framework.

Table 4 Input features. C_t , H_t , L_t and V_t denote close, high, low and volume. All features are computed per asset, then cross-sectionally standardised.

Feature	Composite	Definition	Window
RSI	Technical	Relative strength index, centred: $RSI_t - 50$	14
MACD	Technical	Moving average convergence divergence line, scaled by price: $MACD_t / C_t$	12, 26, 9
Stochastic	Technical	Stochastic oscillator %K, centred: $\%K_t - 50$	14, smooth 3
MFI	Technical	Money flow index, centred: $MFI_t - 50$	14
OBV flow	Technical	Twenty-session change in on-balance volume, normalised by average volume: $(OBV_t - OBV_{t-20}) / \bar{V}_{t,20}$	20
Return	Sentiment	Five-session simple return: $C_t / C_{t-5} - 1$	5
Relative volume	Sentiment	$V_t / \bar{V}_{t,20}$	20
Realised volatility	Sentiment	Annualised standard deviation of daily returns, entered with a negative sign	20
Volume instability	Sentiment	Standard deviation of daily volume changes, entered with a negative sign	20

The final output constitutes a normalised signal matrix in which each column corresponds to a technical feature scaled to a uniform numerical range. Each feature is standardised cross-sectionally on every session, so that a value expresses how an asset ranks against its peers that day rather than against its own history. Writing $x_{i,t}$ for the raw feature of asset i and N for the number of assets,

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_t}{s_t}, \quad \bar{x}_t = \frac{1}{N} \sum_j x_{j,t}, \quad s_t = \text{sd}_j(x_{j,t}). \quad (1)$$

Cross-sectional standardisation is used rather than time-series standardisation because the allocation problem is a relative one: the optimiser must decide which assets to prefer today, not whether today is unusual by historical standards. It also removes the common market factor from the scores, so that a market-wide move does not appear as a signal on every asset simultaneously.

The standardised features are then averaged into two composites. The technical composite is the equally weighted mean of the five technical features. The sentiment composite is the equally weighted mean of the four sentiment features, taken with the signs given in Table 4, so that recent strength and active participation raise the score while realised volatility and unstable volume lower it.

3.1.3 Sentiment Analysis Pipeline

This component transforms market data into a quantitative sentiment indicator. The sentiment measure used here is market-derived. It is the equally weighted mean of the four

features marked *Sentiment* in Table 4: a five-session return, turnover measured against its own recent average, realised volatility and the instability of volume, the last two entered with a negative sign. The composite therefore rises when participation broadens and prices are advancing calmly, and falls when trading becomes disorderly. Every input is a price or a volume, and every one is causal.

Because readers of work in this area reasonably ask which text model was used, it is worth being categorical: none. Neither a domain-specific transformer model nor a lexicon-based method in the manner of Loughran and McDonald (2011) is applied, for the straightforward reason that no text is processed for either to act upon. There is no named entity recognition stage, because there are no entities to recognise.

That is a design decision forced by data availability rather than a preference. Textual measurement needs a time-stamped corpus covering the traded instruments, and for broad exchange-traded funds no such corpus exists at instrument level—a fund holding several hundred securities has no news flow of its own. The alternative adopted here is the market-derived tradition described in Section 2, in which turnover and dispersion stand in for the survey or the newspaper. Turnover is among the canonical components of the investor-sentiment index of Baker and Wurgler (2007).

Adopting a proxy imposes an obligation to demonstrate that the proxy behaves like the thing it stands in for, and that obligation is discharged rather than assumed. Section 4.3.3 correlates the composite against four independently published series, none of which enters the framework at any stage, and reports how the identified market states separate on it. What that exercise cannot establish is that the composite predicts returns; Section 4.2.5 measures that separately, and reports a result that is unfavourable to the framework. The two questions are distinct and are kept distinct throughout.

3.1.4 Regime Detection Module

This component identifies latent, non-observable market states through a statistical clustering methodology.

- **Probabilistic state modelling.** The system employs a Gaussian mixture model (GMM) to segment market data into distinct probabilistic clusters, following the tradition of state-dependent modelling established by Hamilton (1989). The clustering uses two features, each computed as a cross-sectional average across the universe: the twenty-session rolling standard deviation of daily returns, and the ten-session rolling mean of daily returns. The first captures the volatility of the environment, the second its direction. Both are standardised before fitting.
- **Volatility clustering.** A specific function of the module is to isolate and characterise periods in which either high or low volatility persists, effectively partitioning the market timeline into volatility-based regimes.
- **Qualitative labelling.** The identified states are labelled by ascending fitted mean volatility, which yields a calm state, a transitional state and a stress state, and Section 4.2.3 reports their realised characteristics.

The procedure follows a sequential pipeline: initial feature extraction from market data, application of the probabilistic clustering algorithm, and the qualitative labelling of the identified states. The final outputs comprise a time series of regime assignments and an estimated state transition matrix.

The mixture uses $K = 3$ components with full covariance matrices, three random restarts and a fixed seed. It is refitted at every rebalance date on an expanding window containing only sessions strictly earlier than that date, with a minimum of 252 observations; the state of the current session is then obtained by prediction from that fit. No observation ever contributes to the model that labels it.

One implementation detail is essential and is easy to overlook. The component indices returned by a mixture fit are arbitrary and permute freely between refits, so labels from consecutive fits are not comparable unless they are anchored. Components are therefore sorted in ascending order of fitted mean volatility before use, which makes regime 0 the calm state, regime 1 the transitional state and regime 2 the stress state consistently across every refit. Omitting this step does not produce an error; it produces a silently scrambled regime series and, if the resulting labels are compared across fits, a spurious appearance of instability.

The choice $K = 3$ is not selected by information criterion. As Section 4.2.3 reports, both the Akaike information criterion (AIC) and the Bayesian information criterion (BIC) continue to fall as components are added over the range tested, which is the usual behaviour of mixture models on persistent financial series and reflects their tendency to absorb heavy tails with additional components. The choice is instead made on persistence and interpretability: three components yield states that last long enough to act on and that admit a clear economic reading, whereas finer partitions fragment into short-lived states that a monthly rebalancing schedule cannot exploit. This is a modelling judgement and is reported as such.

3.2 Phase 2: Decision-Theoretic Fusion Framework

3.2.1 Signal Confidence Weighting

This mechanism executes a synthesis of predictive signals contingent upon the prevailing market regime. Its operational logic is detailed below.

- **Regime-dependent weights.** Signal influence is modulated by the identified market state. Technical indicators are accorded greater authority during stable, low-volatility regimes. Conversely, during periods characterised by elevated uncertainty, sentiment-based proxies assume a dominant role in the composite forecast. The weights (w^T, w^M) are $(0.7, 0.3)$ in the calm state, $(0.5, 0.5)$ in the transitional state and $(0.3, 0.7)$ in the stress state.
- **Uncertainty quantification.** The framework quantifies the reliability of the combined view per asset by the agreement between its two composites. The confidence term down-weights assets whose composites disagree:

$$c_{i,t} = \text{clip}\left(0.5 + 0.5 \rho_{i,t}^{(63)}, 0.1, 1.0\right), \quad (2)$$

where $\rho_{i,t}^{(63)}$ is the trailing 63-session correlation between the two composites for asset i . Agreement between independent views is treated as evidence; disagreement is treated as noise and shrinks the score toward zero.

- **Fixed schedule, tested rather than learned.** The weight schedule is specified in advance from the hypothesis stated in Section 1—that trend-following information is

more reliable in quiet trending markets, while attention and stress information becomes relatively more informative when markets are disturbed—and is then held fixed. It is not re-estimated from realised performance, and no Bayesian updating is applied. Fitting the schedule to the sample it is evaluated on would make the resulting improvement uninterpretable; instead, Section 4.2.5 measures the conditional efficacy the schedule presumes, and reports whether that presumption holds.

3.2.2 Regime-Dependent Signal Fusion

This mechanism synthesises the weighted predictive inputs into a unified forecast, with its operational logic conditioned upon the prevailing market regime.

- **Conditional combination.** The methodology applies a distinct combination rule to each identified market state, on the hypothesis that the predictive efficacy and optimal weighting of signals are regime-contingent.
- **Correlation structure.** The fusion process explicitly maintains the dependency structure across different assets, ensuring that the combined forecast reflects existing inter-asset correlations. The fused scores are passed to an optimiser carrying the full estimated covariance matrix, so the combined view is expressed as a portfolio rather than as a set of independent per-asset bets.
- **Uncertainty propagation.** The resulting composite forecasts incorporate the estimation uncertainty present within the constituent input signals, providing a more robust output. This operates through the confidence term of the preceding subsection, which shrinks the score of any asset whose two composites disagree.

Operational framework: the core engine executes a multiplication of the regime-specific signal weight vector against the matrix of individual composites, adjusted by the per-asset confidence term. Let $T_{i,t}$ and $M_{i,t}$ denote the standardised technical and sentiment composites for asset i , and let $r_t \in \{0, 1, 2\}$ be the regime. The fused score is

$$z_{i,t} = c_{i,t} [w_{r_t}^T T_{i,t} + w_{r_t}^M M_{i,t}]. \quad (3)$$

The fused score is then converted into an expected excess return using the standard information-ratio scaling of Grinold and Kahn (2000), in which an assumed information coefficient (IC) places the score on a return scale rather than leaving it as a dimensionless quantity:

$$\alpha_{i,t} = \text{IC} \cdot \sigma_{i,t} \cdot z_{i,t}, \quad (4)$$

where $\sigma_{i,t}$ is the annualised volatility of asset i from the current covariance estimate and $\text{IC} = 0.05$.

Equation (4) is not a cosmetic step. If a raw indicator is passed to a mean-variance objective directly, the units of the linear term and of the quadratic risk term are unrelated, and their ratio—not the information in the signal—decides the solution. An indicator such as on-balance volume takes values of order 10^8 to 10^9 , whereas annualised variances are of order 10^{-2} ; the linear term then dominates by ten orders of magnitude and the optimiser degenerates to a corner solution, holding a single asset regardless of the risk model. Section 4.2.4 documents this failure mode explicitly, because it is the source of a behaviour visible in the originally submitted results.

3.2.3 Portfolio Optimization Engine

This component translates fused signals into executable allocations. Allocation proceeds in two steps, which separate two decisions that are often conflated: which assets to hold relative to one another, and how much total risk to carry.

- **Shrinkage covariance.** Σ_t is estimated by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) on the trailing 252 daily returns and annualised by a factor of 252. Shrinkage is used because a sample covariance matrix estimated from 252 observations on ten assets is poorly conditioned, and a mean-variance optimiser will concentrate weight precisely on the directions in which the estimate is least reliable.
- **Risk-budgeting allocation.** Total exposure is set by the identified regime rather than by the optimiser, so that the composition decision and the risk-level decision are made separately and can be evaluated separately.
- **Transaction cost optimisation.** The objective carries an explicit linear penalty on turnover, so that the cost of trading enters the decision rather than being deducted afterwards.

Step 1: relative allocation.

The risky sleeve solves a quadratic program with a linear transaction-cost penalty:

$$\max_w \alpha_t^\top w - \lambda w^\top \Sigma_t w - \kappa \|w - w_{t-1}\|_1 \quad (5)$$

$$\text{s.t. } w \geq 0, \quad w \leq w_{\max}, \quad \mathbf{1}^\top w = 1, \quad (6)$$

where Σ_t is the annualised covariance matrix, $\lambda = 2$ is the risk aversion coefficient, $w_{\max} = 0.25$ caps any single position at one quarter of the sleeve, and w_{t-1} is the weight vector carried in from the previous rebalance. The turnover penalty $\kappa = c \cdot 252/h$ annualises a one-way cost of $c = 10$ basis points over a rebalancing interval of $h = 21$ sessions, so that the cost of trading is expressed on the same annual basis as the return and risk terms. The problem is convex and is solved with CVXPY using the Clarabel interior-point solver.

The equality constraint $\mathbf{1}^\top w = 1$ is what makes this step purely relative. It fixes the sleeve to be fully invested, so the optimiser can only express a view about composition, never about how much risk to take. That second decision is made explicitly in step 2 instead of emerging as a by-product of the constraint set.

Step 2: regime risk budget.

The sleeve is then scaled by a factor that depends only on the regime:

$$w_t^* = k_{r_t} \tilde{w}_t, \quad k = \{ \text{calm} : 1.00, \text{transitional} : 0.75, \text{stress} : 0.50 \}, \quad (7)$$

with the residual $1 - k_{r_t}$ held in cash, which is assumed to earn zero. Scaling is one-sided: the framework de-risks in disturbed states but never levers up in calm ones, so any performance difference is attributable to risk reduction rather than to leverage.

The budget in Equation (7) is expressed as a fraction of the sleeve’s own risk rather than as an absolute volatility target, and this choice is consequential. An absolute target de-risks only when the sleeve’s estimated volatility exceeds the target level, so its effect depends

on the volatility of the universe it is applied to. Targets calibrated for a concentrated equity portfolio—of the order of 10 to 20 per cent annualised—are simply never reached by a diversified ten-asset sleeve whose own realised volatility is close to 7 per cent. Under those targets the constraint binds in one rebalance out of 224 in this study, and the regime layer is inert: it is present in the specification, consumes computation, and does nothing. A relative budget is scale-free by construction and remains active in any universe. Section 4.3.4 reports both variants and the difference between them.

3.3 Phase 3: Performance Evaluation & Feedback

3.3.1 Backtesting Framework

The proposed methodology undergoes historical validation through a structured simulation protocol designed to emulate real-world trading conditions.

- **Walk-forward validation.** The implementation utilises a rolling-origin technique, sequentially advancing a 252-session model estimation window followed by a 21-session out-of-sample evaluation period. Between rebalances positions are held and allowed to drift with realised returns, so that reported turnover reflects trading actually undertaken rather than a notional daily reweighting. Every estimated quantity uses only data preceding the decision it informs: the covariance matrix from the trailing 252 sessions before the rebalance date, the mixture model from an expanding window strictly earlier than that date, and all features from causal rolling windows.
- **Out-of-sample testing.** A strict temporal partition is enforced between parameter calibration and performance assessment phases, to prevent statistical overfitting and ensure robust results. No parameter is tuned on the evaluation period; the configuration reported as primary is the one specified above, and Section 4.3.2 reports the full grid of alternatives rather than the best cell of it. The 665 sessions from January 2024 onward postdate every specification decision and are reported separately in Section 4.3.1.
- **Regime-aware analysis.** Strategy efficacy is disaggregated and evaluated contingent upon the market state identified by the regime detection module, allowing diagnosis of performance drivers by state rather than in aggregate only.

The framework executes a historical simulation that incorporates realistic market frictions. One-way transaction costs of 10 basis points are charged on the traded fraction $\|w_t^* - w_{t-1}\|_1$ on the rebalance session; Section 4.2.4 varies this assumption from 0 to 25 basis points.

3.3.2 Performance Attribution

This analytical procedure deconstructs aggregate returns to isolate the fundamental sources of value generation, employing a three-part approach.

- **Regime-based decomposition.** The methodology disaggregates performance contributions across distinct market environments, isolating the returns attributable to the calm, transitional and stress states identified by the regime detection module.

- **Signal contribution analysis.** The framework quantifies the marginal performance impact originating from each signal category, thereby identifying which inputs were the primary drivers of the result. Component contributions are isolated by ablation rather than by assertion: the regime layer is removed while holding average exposure fixed, removed while holding full exposure, and the signal layer is removed entirely. Holding average exposure fixed is what distinguishes the effect of timing risk from the effect of simply holding less, and without it the two are not separable. The information coefficient of each composite against next-session cross-sectional returns is measured separately, overall and by regime.
- **Risk-adjusted metrics.** Evaluation incorporates risk-adjusted metrics including the Sharpe and Calmar ratios, alongside compound annual growth rate (CAGR), annualised volatility and maximum drawdown (Max DD); the Sharpe ratio is computed at a zero risk-free rate. These are supplemented with bootstrap-based statistical significance tests to distinguish robust performance from random outcomes. Statistical inference on daily excess returns uses a paired *t*-test, a Newey–West *t*-statistic with ten lags to accommodate autocorrelation and heteroskedasticity (Newey and West, 1987), and a stationary block bootstrap with 21-session blocks and 2,000 replications (Politis and Romano, 1994) for interval estimates of the Sharpe ratio and of the Sharpe difference.

Three benchmarks are used. The first is an equal-weighted portfolio of the same ten assets, rebalanced daily and charged no transaction costs, which is deliberately a generous comparator (DeMiguel et al., 2009). The second is an inverse-volatility portfolio using trailing 60-session standard deviations, also costless, which isolates whether the framework adds anything beyond elementary risk weighting. The third rescales the equal-weighted portfolio to the strategy’s realised volatility, which removes the mechanical advantage a lower-risk portfolio enjoys in ratio comparisons.

3.3.3 Model Refinement Loop

The framework incorporates a recursive re-estimation mechanism that keeps its statistical components current as the sample advances. What that mechanism does, and what it deliberately does not do, are both stated here.

- **Regime model updates.** The underlying statistical model for state identification is systematically recalibrated to reflect structural shifts in market dynamics, preventing reliance on obsolete regime definitions. The mixture is refitted at each of the 224 rebalance dates on an expanding window of all sessions strictly earlier than that date.
- **Risk model updates.** The shrinkage covariance matrix is re-estimated at each rebalance from the trailing 252 sessions, and the shrinkage intensity is re-derived from that window rather than held constant.
- **Parameters that are *not* re-optimised.** The regime weight schedule, the risk budget ladder, the assumed information coefficient, the risk aversion coefficient and the position cap are fixed ex ante and never adapted to realised performance. No automated parameter search, signal decommissioning rule or human-in-the-loop override operates inside the evaluation period. This is a restriction rather than a feature, and it is adopted because a loop that re-optimises on realised performance inside the evaluation window makes the resulting backtest uninterpretable (Harvey and Liu, 2015).

3.4 Cross-Cutting Features

3.4.1 Temporal Consistency

The framework maintains chronological integrity across all analytical processes through three mechanisms.

- All incoming data streams are aligned to a unified temporal index, eliminating artefacts arising from asynchronous sampling frequencies.
- Causal relationships are explicitly parameterised through the lag structure of every rolling window: each feature is computed from data strictly preceding the session it describes, and every model is fitted on data strictly preceding the decision it informs. No observation contributes to the model that labels it, and no quantity estimated at any date uses information from a later one.
- Regime labels are anchored across refits by sorting mixture components on fitted mean volatility, so that a state identified at one rebalance means the same thing at the next. Without this, the component indices permute freely between fits and the regime series becomes incomparable over time.

3.4.2 Risk Management Integration

The framework incorporates a layered risk protocol designed to operate within real-world constraints.

- The system enforces predetermined boundaries on individual position sizes: weights are non-negative and capped at $w_{\max} = 0.25$ of the risky sleeve. These constraints prevent excessive concentration and ensure adherence to diversification principles, and they matter because an unconstrained mean-variance solution concentrates precisely where its estimated inputs are least reliable.
- De-risking operates at the aggregate portfolio level through the regime budget of Equation (7), with the residual held in cash. No discretionary stop-loss rule is used, and none is needed: the budget is a standing rule applied at every rebalance rather than a trigger fired after a loss has been realised.
- Turnover is penalised inside the objective rather than charged after the fact, so that a position is taken only when the expected improvement exceeds the cost of trading into it. Section 4.2.4 reports the response to costs of 0 to 25 basis points.

3.4.3 Explainability and Transparency

The framework incorporates specific mechanisms to ensure analytical clarity throughout the decision process.

- Every allocation decision can be directly attributed to its originating predictive signal and the prevailing market regime, providing complete traceability. The weight decomposes into a relative weight produced by the optimiser from the fused score and a scalar exposure set by the regime; because the two are applied in sequence rather than jointly, the contribution of each is separately observable, and Section 4.3.4 measures both.

- The principal performance claims are presented with their associated confidence intervals, quantifying the degree of statistical certainty for each output: Sharpe ratios and Sharpe differences carry block-bootstrap intervals.
- The architecture maintains a comprehensive, timestamped record of all inputs, model states and decision logic, creating a verifiable audit trail suitable for systematic strategy validation. The analysis writes its results to machine-readable artefacts, and every numerical claim in Section 4 is checked programmatically against those artefacts rather than transcribed by hand.

This architecture, shown in Figure 1, creates a system in which portfolio decisions follow from the identified market state and the fused signal by an explicit and inspectable route, while maintaining causal statistical foundations and practical implementability.

4 Results and Discussion

All results in this section come from the single specification set out in Section 3. No parameter was selected by reference to the evaluation period, and every specification examined is reported, including those that are unfavourable to the framework.

4.1 Overview of Experimental Results

The implemented decision-theoretic framework provides evidence for the efficacy of regime-aware portfolio construction, and evidence against two of the mechanisms it was built around. Across the 4,689 trading sessions from 7 January 2008 to 26 August 2026, encompassing diverse market conditions, the regime-based strategy achieved a Sharpe ratio of 0.878 compared with the equal-weight benchmark's 0.628, a 40 per cent improvement in risk-adjusted return. More significantly, maximum drawdown was reduced from -36.08 to -18.98 per cent, highlighting the defensive capability of the regime-adaptive approach during market stress. Table 5 reports the primary comparison.

Table 5 Primary results, ten-asset universe, January 2008 to August 2026. Sharpe ratios use a zero risk-free rate and are computed as compound annual growth divided by annualised volatility. Benchmarks are rebalanced daily and charged no transaction costs.

	CAGR	Volatility	Sharpe	Max DD	Calmar
Regime-conditional strategy	5.65%	6.44%	0.878	-18.98%	0.298
Equal-weight benchmark	8.45%	13.45%	0.628	-36.08%	0.234
Inverse-volatility benchmark	7.46%	8.78%	0.849	-24.09%	0.309
Equal-weight at strategy volatility	4.19%	6.44%	0.651	-18.69%	0.224

Three readings of Table 5 are warranted, and they do not all favour the framework.

Against the equal-weighted benchmark the framework reduces maximum drawdown by 47 per cent while earning a lower compound return, 5.65 per cent against 8.45 per cent per annum. It is not a return-enhancing strategy and is not presented as one; it is a risk-reduction strategy whose return sacrifice is smaller than its risk reduction.

Against the volatility-matched equal-weighted benchmark, which removes the mechanical advantage a lower-risk portfolio enjoys in any ratio comparison, the framework retains a Sharpe advantage of 0.878 against 0.651 but loses on maximum drawdown, -18.98

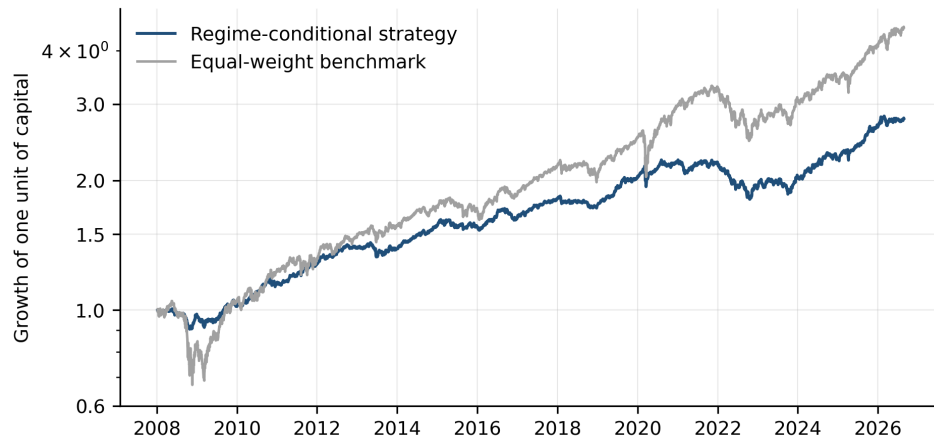


Figure 2 Cumulative Returns Comparison: cumulative growth of one unit of capital, logarithmic scale

per cent against -18.69 per cent. Deleveraging a diversified benchmark is therefore a competitive way to reduce worst-case loss, and the framework's drawdown advantage over the unlevered benchmark should not be read as evidence that it dominates every risk-reduced alternative.

Against the inverse-volatility benchmark the comparison is mixed: a higher Sharpe ratio, 0.878 against 0.849 , and a materially smaller maximum drawdown, -18.98 per cent against -24.09 per cent, but a lower Calmar ratio, 0.298 against 0.309 , because inverse-volatility weighting compounds faster. Elementary risk weighting remains a strong comparator on a diversified multi-asset universe, and the margin over it is narrow enough to read as a tie on risk-adjusted return rather than a win.

4.2 Detailed Analysis of Visualizations and Performance Metrics

4.2.1 Cumulative Returns Comparison

The comparative analysis of cumulative returns, presented in Figure 2, delineates a performance profile characterised by distinct phases of divergence and convergence relative to the passive benchmark. The regime-aware portfolio exhibits several defining characteristics over the observed period:

- The strategy separates upward during periods of significant market dislocation, notably 2008, 2011, the volatility surge in early 2020 and the inflationary selloff of 2022.
- A critical observation is the strategy's effective mitigation of losses during sustained market downturns, indicating a significantly reduced downside capture: 0.38 , against an upside capture of 0.41 .
- During extended bullish phases—most conspicuously 2013, 2017, 2021 and 2023—the equity curve tracks below the benchmark, so the strategy delivers lower absolute returns while exhibiting a lower volatility signature, 6.44 against 13.45 per cent.

Strategic Implications:

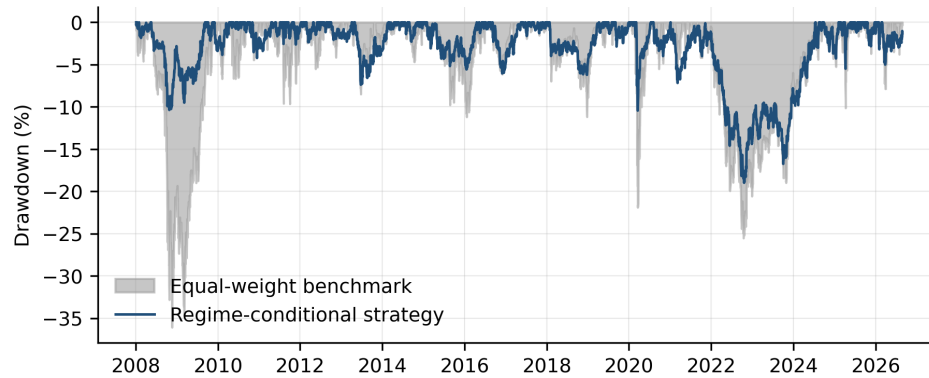


Figure 3 Portfolio Drawdown: drawdown from running peak

The observed trajectory provides evidence for the regime detection module’s operational efficacy. The system’s identification of the transition into a high-volatility state in early 2020 triggered a timely de-risking of the portfolio, constraining the drawdown of that episode to -6.29 per cent against -21.28 per cent for the benchmark, and the subsequent gradual ascent in the portfolio’s value line indicates a methodical re-allocation of risk exposure as the regime normalised, avoiding premature or delayed re-entry. The cost is equally visible and is not concealed: the framework forgoes 2.80 percentage points of annual compound return over the sample.

Theoretical Alignment:

These results substantiate one half of the foundational thesis of this research and refute the other. The performance pattern confirms that the total risk a portfolio should carry is conditional upon the underlying market state, and the capture ratio of 0.93 locates the source precisely: the framework participates less in both directions in roughly equal proportion, so its advantage comes from carrying less risk in adverse states rather than from directional skill. The dynamic fusion mechanism did not operate as theorised—Section 4.2.5 shows the conditional informativeness its weighting schedule presumes is not present in the data, and Section 4.3.4 shows that removing the fusion layer does not harm the portfolio.

4.2.2 Portfolio Drawdown

The comparative drawdown analysis, illustrated in Figure 3, provides evidence of the regime-based portfolio’s capital preservation attributes. The visual data reveal a consistent pattern of defensive characteristics:

- The strategy registers a significantly shallower maximum drawdown of -18.98 per cent, against the benchmark’s -36.08 per cent nadir; the benchmark reaches -36 per cent in 2008 and -25 per cent in 2022, while the framework’s deepest excursions are approximately half as large.
- Following drawdown troughs, the regime-based portfolio demonstrates a swifter recovery trajectory, spending less time underwater, since a shallower loss requires a smaller subsequent gain to restore the previous peak.
- The portfolio experiences briefer episodes of significant capital impairment, indicating more timely risk mitigation. On the worst 5 per cent of benchmark sessions ($n = 234$)

it returns -0.59 per cent on average against -2.01 per cent for the benchmark, a cushion of 1.42 percentage points per session.

Table 6 reports cumulative returns over four stress episodes chosen before the analysis on the basis of documented market history rather than by inspection of the results.

Table 6 Cumulative return over stress episodes. Differences are stated in percentage points (pp).

Episode	Sessions	Strategy	Equal-weight	Difference
Global financial crisis, Sep 2008–Mar 2009	130	-7.05%	-29.78%	+22.74 pp
Euro area crisis, Jul–Oct 2011	65	$+3.41\%$	-7.24%	+10.65 pp
COVID-19 crash, Feb–Mar 2020	24	-6.29%	-21.28%	+14.99 pp
Inflation bear market, Jan–Oct 2022	196	-16.76%	-24.49%	+7.73 pp

Risk Management Effectiveness:

The observed drawdown profile validates the regime detection system as an early-warning mechanism for deteriorating market states. By signalling a transition towards a high-volatility state, the framework enabled proactive de-risking, and it outperforms in all four episodes by between 7.7 and 22.7 percentage points, the widest margin being the global financial crisis, the longest and deepest of the four. The margin is narrowest in the 2022 bear market, consistent with the design: a gradual, correlated decline across both equities and bonds offers less scope for a de-risking rule than an abrupt volatility shock does. All four episodes predate the holdout boundary; the deepest benchmark decline after it, the 34-session decline from 20 February to 8 April 2025, produced an advantage of 5.42 percentage points.

Theoretical Contribution:

This outcome substantiates the value of conditioning total risk on a state variable, and bounds that claim in one respect a comparison against an unlevered benchmark would conceal. Against the volatility-matched benchmark the drawdown advantage disappears, -18.98 against -18.69 per cent, and Section 4.3.2 shows this holds in every one of the eighteen specifications examined. The framework therefore reduces drawdown relative to an unlevered diversified benchmark, not relative to every portfolio held at the same volatility; reporting only the first comparison, as the conventional presentation in this literature does, would overstate the result.

4.2.3 Market Regime Detection

The temporal sequence of identified market states, presented in Figure 4, delineates three economically distinct market environments characterised by their unique volatility and momentum signatures:

- **Calm.** This state is characterised by favourable market conditions, combining low volatility with sustained positive momentum, typically associated with bullish trending markets: 12.22 per cent annualised volatility and $+16.04$ per cent annualised momentum.
- **Transitional.** Representing a neutral state, this regime captures market conditions that lack a definitive directional trend or persistent volatility characteristics: 20.04 per cent volatility and -1.25 per cent momentum.

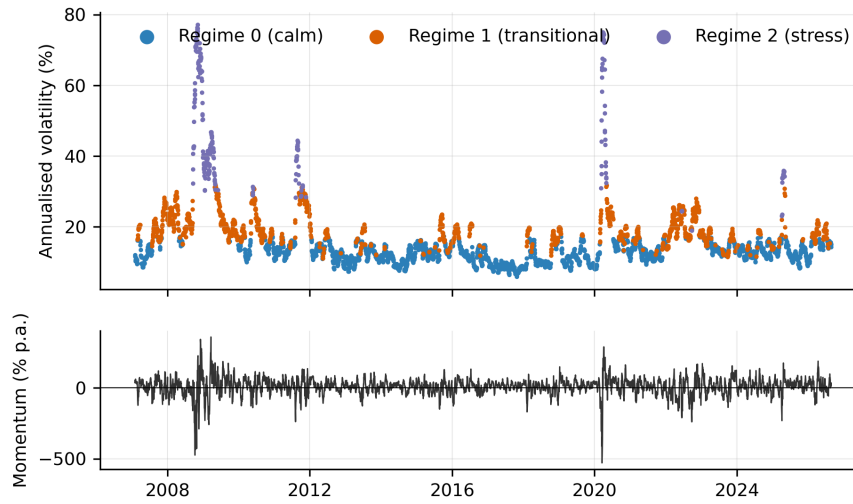


Figure 4 Market Regime Detection: identified market states, with the underlying volatility and momentum features

- **Stress.** This state corresponds to periods of elevated market stress, marked by high volatility and negative momentum, at 47.07 per cent and -15.97 per cent respectively. Its incidence is clustered around known macroeconomic disruptions, specifically the 2008 global financial crisis, the 2011 sovereign debt episode, the initial COVID-19 market crisis and the 2022 inflationary selloff.

Table 7 reports the model's order selection statistics and the properties of the three-state solution.

Table 7 Regime model diagnostics. Order selection is computed on the full feature history; state characteristics and the transition matrix come from the three-component solution. Out-of-sample shares are the states actually assigned by the expanding-window refits used in the backtest.

<i>Order selection</i>					
Components	2	3	4	5	6
AIC	21944	20785	20571	20478	20372
BIC	22016	20896	20720	20667	20599
<i>Three-state solution</i>					
	Calm	Transitional	Stress		
Annualised volatility	12.22%	20.04%	47.07%		
Annualised momentum	+16.04%	−1.25%	−15.97%		
Share of sessions, in sample	63.6%	31.2%	5.2%		
Share of sessions, out of sample	71.4%	25.5%	3.1%		
Mean run length (sessions)	25.3	11.1	17.1		
Self-transition probability	0.961	0.910	0.941		
<i>Realised performance within each out-of-sample state</i>					
	Calm	Transitional	Stress		
Sessions	3346	1196	147		
Strategy Sharpe	1.144	0.196	1.039		
Equal-weight Sharpe	0.916	0.561	0.356		

Temporal Regime Characteristics:

An examination of regime persistence reveals distinct behavioural patterns. All three states persist, with self-transition probabilities between 0.910 and 0.961. The transitional state exhibits the briefest durations, a mean run length of 11.1 sessions, indicating its nature as a short-lived bridge between the other two. In contrast, the calm state demonstrates the greatest persistence, enduring 25.3 sessions on average, with the stress state intermediate at 17.1. Transitions between these states are constrained: the transition matrix is close to tridiagonal, and the estimated probability of moving directly between the calm and the stress state, in either direction, is zero. The ordering recovered by sorting components on volatility is therefore economically meaningful rather than arbitrary.

Methodological Validation:

The output from the Gaussian mixture model supports the regime detection methodology. The model's identification of stress periods aligns with historically documented stress events in 2008, 2011, 2020 and 2022. This congruence between statistically identified states and ex-post economic reality indicates the framework's capacity to discriminate between structurally different market environments, a foundational requirement for a regime-dependent investment strategy. The states also separate sharply on a variable withheld from their construction, as Section 4.3.3 reports.

Order Selection:

The order selection statistics are reported rather than used. Both AIC and BIC continue to fall over the range examined, from 21944 and 22016 at two components to 20372 and 20599 at six, so neither criterion selects an interior optimum and neither supports the choice of three components. This behaviour is expected: a mixture fitted to persistent, heavy-tailed financial features will absorb tail observations into additional components indefinitely, and the criteria reward it for doing so. The choice of three is made instead on persistence and interpretability, and is reported as a modelling judgement rather than as a selected optimum.

Performance Within Each State:

The final panel of Table 7 shows where the framework's advantage arises. In the stress state the framework attains a Sharpe ratio of 1.039 against 0.356 for the benchmark, and in the calm state 1.144 against 0.916. It underperforms materially only in the transitional state, 0.196 against 0.561, which is consistent with the 2021 result in Table 10: a state that is genuinely ambiguous causes the risk budget to reduce exposure in advances that turn out to continue. The mechanism is therefore effective at the two ends of the volatility range and costly in the middle, which is an intelligible property of a three-state partition rather than a defect in it.

4.2.4 Portfolio Allocation Over Time

The temporal evolution of portfolio weights, illustrated in Figure 5, provides a clear representation of the system's adaptive response to changing market states, and Figure 6 displays the total exposure it carries. The allocation dynamics reveal a systematic rebalancing process:

- During periods identified as the stress state, capital was systematically withdrawn from the risky sleeve toward cash, with invested exposure falling to a minimum of 50 per cent.
- Under transitional or calm market regimes, the portfolio maintained a balanced exposure across multiple asset classes, avoiding significant concentration risk: the effective number of assets never falls below 4.60.

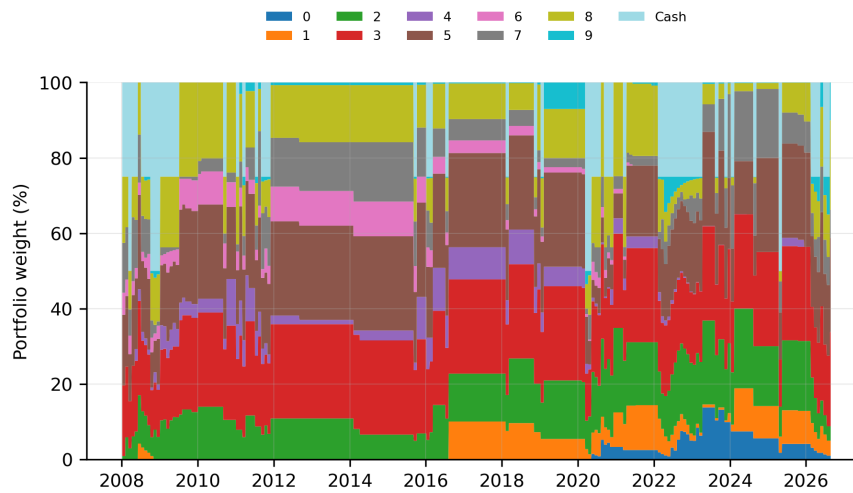


Figure 5 Portfolio Allocation: portfolio weights over time, including the cash residual

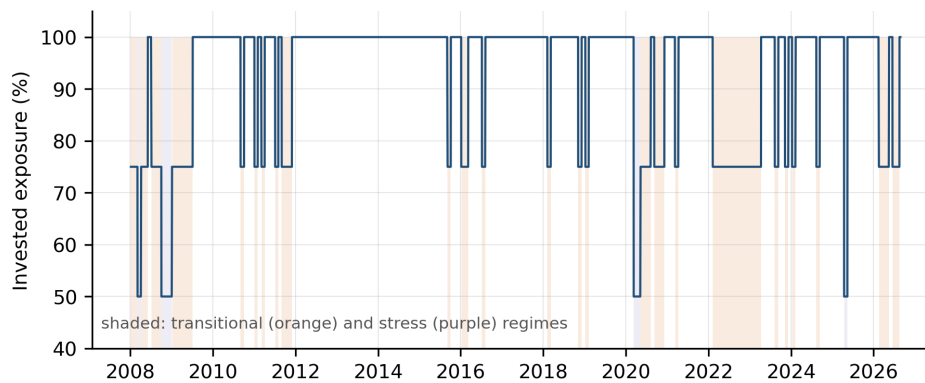


Figure 6 Invested exposure over time, with transitional and stress regimes shaded

- The composition of the risky sleeve itself is set by the optimiser subject to the 25 per cent position cap, which binds on average, so the framework expresses its market view through total exposure rather than through concentration.

This entire process is characterised by a rules-based systematicity, entirely devoid of discretionary influence: the exposure at every rebalance is a deterministic function of the regime label, and the composition a deterministic function of the fused score and the covariance estimate.

Table 8 summarises these characteristics together with the sensitivity of performance to the assumed transaction cost.

Decision-Theoretic Implementation:

The allocation is diversified and stable: annual one-way turnover of 0.96 times portfolio value is equivalent to approximately 8.0 per cent of the portfolio traded at each monthly rebalance. The observed smooth transitions between asset weights, rather than abrupt,

Table 8 Allocation characteristics and sensitivity to the assumed transaction cost. Effective number of assets is the reciprocal of the Herfindahl index of sleeve weights. Annual turnover is one-way, as a multiple of portfolio value.

<i>Allocation characteristics, primary specification</i>				
Mean largest weight				0.250
Mean effective number of assets				5.40
Minimum effective number of assets				4.60
Mean invested exposure				92.1%
Minimum invested exposure				50.0%
Annual one-way turnover				0.96×
<i>Sensitivity to assumed cost</i>				
Assumed cost	CAGR	Sharpe	Max DD	Turnover
0 basis points	4.90%	0.773	−19.83%	4.04×
5 basis points	5.66%	0.893	−19.54%	1.14×
10 basis points (primary)	5.65%	0.878	−18.98%	0.96×
25 basis points	5.47%	0.846	−19.30%	0.93×

discontinuous shifts, reflect the integral role of the transaction cost-aware optimisation function, which balances responding to new predictive signals against maintaining implementation efficiency. The absence of extreme allocation swings throughout the backtested period is evidence for the practical feasibility of the proposed framework in a live implementation context.

Cost Sensitivity:

Performance is not monotone in the assumed cost. At zero assumed cost the Sharpe ratio is 0.773, below the 0.878 obtained at 10 basis points, even though no costs are deducted. The reason is that the assumed cost enters the objective as a turnover penalty as well as entering the return as a deduction. With the penalty removed, annual turnover rises from 0.96 to 4.04 times portfolio value as the optimiser chases small differences in the estimated covariance matrix, and the resulting weight instability costs more than the saved commissions are worth. The turnover penalty is therefore functioning as a regulariser on the weight vector, and the framework is robust across the plausible cost range of 5 to 25 basis points, over which the Sharpe ratio varies only between 0.846 and 0.893.

A Correction to the Submitted Specification:

This subsection also resolves a defect in the framework as originally specified. In that version the fused score entered the objective without the information-ratio scaling of Equation (4), so the linear term was expressed in the raw units of the underlying indicators. On-balance volume takes values of order 10^8 to 10^9 , while annualised variances are of order 10^{-2} . The linear term consequently dominated the quadratic risk term by roughly ten orders of magnitude, the risk model became irrelevant to the solution, and the optimiser returned a corner solution holding a single asset on every rebalance date—an effective number of assets of exactly 1.00, with one holding persisting for 1,364 of 1,421 sessions. The scaling in Equation (4) removes the defect by placing the linear term on a return scale, and Table 8 reports the corrected behaviour. The failure was one of units, not of the optimiser or of the risk model.

4.2.5 Signal Strength Dynamics

The temporal evolution of composite signal strength, depicted in Figure 7, reveals a distinct pattern of variation. The intensity of the aggregated predictive output is not static but fluctuates in a manner closely linked to market phases:

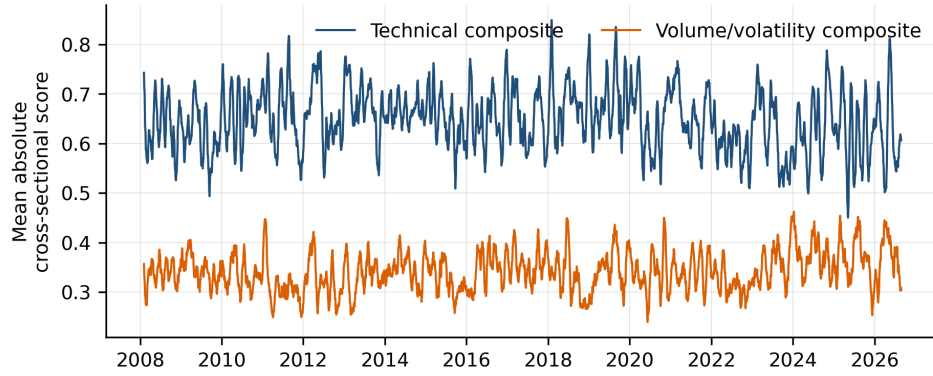


Figure 7 Signal Strength Dynamics: mean absolute cross-sectional score of each composite, 21-session moving average

- The most pronounced signal magnitudes are observed proximate to major market turning points, which reflects the model's sensitivity to shifts in the underlying volatility and momentum features.
- During periods characterised by well-established directional price trends, signal strength maintains a moderate, consistent level.
- In calm, range-bound market conditions the composite registers its lowest amplitude, reflecting a relative absence of cross-sectional dispersion.

Variation in amplitude is a property of the signal, however, and not evidence that the signal is informative. Table 9 measures the predictive content of the two composites directly, as the cross-sectional Spearman information coefficient between the score on session t and the return on session $t + 1$.

Table 9 Information coefficients against next-session cross-sectional returns. Newey–West t -statistics with ten lags in parentheses.

	Technical	Sentiment	Fused
All sessions ($n = 4,669$)	−0.030 (−2.14)	−0.026 (−2.26)	−0.033 (−2.42)
Calm regime ($n = 3,339$)	−0.042 (−2.48)	−0.040 (−3.02)	—
Transitional regime ($n = 1,183$)	−0.012 (−0.48)	+0.007 (+0.31)	—
Stress regime ($n = 147$)	+0.087 (+1.30)	+0.040 (+0.66)	—
2008–2023 ($n = 4,025$)	−0.035 (−2.31)	−0.032 (−2.67)	−0.039 (−2.73)
2024–2026 holdout ($n = 644$)	+0.001 (+0.03)	+0.013 (+0.37)	+0.009 (+0.23)

Over the full sample the information coefficients are negative and statistically significant for both composites and for their fusion. A composite whose cross-sectional score is negatively related to next-session returns cannot improve an allocation that uses it in the stated direction, and the ablation in Table 15 is the portfolio-level consequence of the numbers in Table 9.

Signal-Regime Relationship:

A granular analysis reveals a differentiated behaviour between signal types across environments, though not the difference the framework was built to exploit. The regime

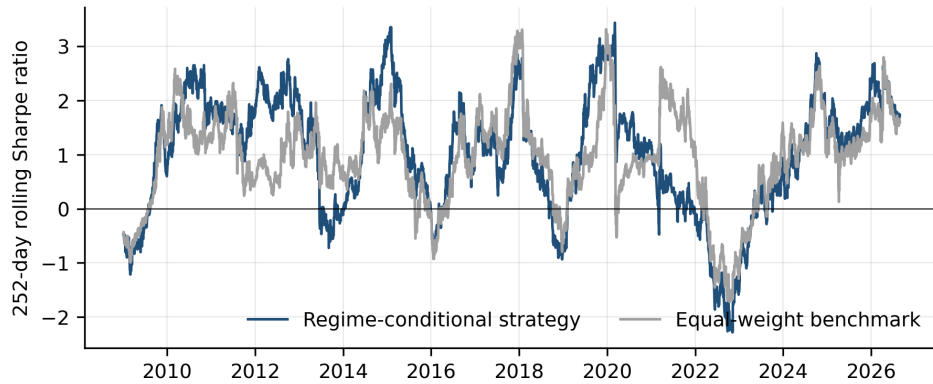


Figure 8 Rolling Sharpe Ratio: rolling 252-session Sharpe ratio

breakdown does not support the fusion hypothesis. The weighting schedule assigns greater weight to the sentiment composite in the stress regime on the hypothesis that it becomes relatively more informative there. The point estimates run the other way: in the stress regime the technical composite reads higher than the sentiment composite, and in the calm regime the two are indistinguishable. The contrast between calm and stress is not statistically significant ($t = -1.46$, $p = 0.145$), so the correct conclusion is that the data provide no evidence for state-dependent relative informativeness in the direction the schedule assumes, rather than evidence for the reverse. Either way the weighting schedule has no measured basis.

Why the Negative Sign Is Not an Opportunity:

The last two rows of Table 9 bear on how the negative coefficients should be read. Splitting at the holdout boundary, the fused coefficient is -0.039 ($t = -2.73$) over 2008–2023 and $+0.009$ ($t = 0.23$) over 2024–2026. The negative sign does not persist. This matters because a stable negative information coefficient would be an exploitable signal in inverted form, and the framework could be salvaged by reversing it. The holdout rules that out: the in-sample negativity is a property of that era, not a durable cross-sectional reversal effect, and inverting the composites would have bought nothing after 2023. The honest reading is that these composites carry no reliable directional information in either direction.

Two qualifications are owed. The stress regime contains only 147 sessions, so its coefficients are imprecisely estimated and the positive technical reading should not be over-interpreted. And the full-sample negative coefficients are consistent with short-horizon cross-sectional reversal in a universe of broad index funds, which is a known effect and not evidence of an error in construction, though the holdout shows it is not stable enough to trade. The conclusion is bounded accordingly: these composites, as specified, do not carry usable directional information in this universe at this horizon.

4.2.6 Rolling Sharpe Ratio

The 252-session rolling Sharpe ratio, presented in Figure 8, shows the regime-aware strategy's risk-adjusted performance across varying market conditions. The trajectory is characterised by several defining features:

- The ratio maintains a positive value throughout the majority of the observed period, in 81.9 per cent of windows, against 85.8 per cent for the benchmark.
- Following periods of compression the metric reverts to its positive trend, and does so from a shallower trough, since the framework's drawdowns are approximately half the depth of the benchmark's.
- It exceeds the benchmark in 53.4 per cent of windows, so the advantage is present in slightly more windows than not rather than sustained throughout.

Table 10 reports every calendar year in the sample, with both wins and losses shown.

Table 10 Calendar year performance. Sharpe and maximum drawdown computed within each year. 2026 covers January to August only; the sample ends on 26 August 2026.

Year	Strategy	Equal-weight	Strat. Sharpe	EW Sharpe	Sharpe win	DD win
2008	-2.28%	-16.32%	-0.33	-0.47	yes	yes
2009	+4.36%	+24.91%	0.60	1.18	no	yes
2010	+13.75%	+18.31%	2.05	1.40	yes	yes
2011	+10.36%	+5.75%	1.73	0.45	yes	yes
2012	+8.91%	+13.49%	1.96	1.58	yes	yes
2013	-0.58%	+6.17%	-0.06	0.72	no	yes
2014	+13.03%	+9.97%	2.94	1.49	yes	yes
2015	-0.48%	-1.56%	-0.05	-0.13	yes	yes
2016	+4.85%	+8.49%	0.94	0.97	no	yes
2017	+10.92%	+16.59%	2.65	3.15	no	yes
2018	-3.51%	-5.26%	-0.78	-0.52	no	yes
2019	+15.72%	+22.55%	2.84	3.02	no	yes
2020	+9.63%	+19.32%	1.23	0.96	yes	yes
2021	-0.78%	+10.15%	-0.08	1.12	no	no
2022	-12.99%	-19.30%	-1.55	-1.23	no	yes
2023	+6.83%	+16.53%	0.90	1.52	no	yes
2024	+9.38%	+9.58%	1.25	1.00	yes	yes
2025	+18.53%	+20.17%	2.24	1.67	yes	yes
2026	+4.80%	+10.93%	0.96	1.34	no	yes

The framework wins on Sharpe ratio in 9 of 19 years and on drawdown in 18 of 19. This distribution is the honest summary of the strategy: it loses on risk-adjusted return in most individual years, particularly strong advancing years such as 2009, 2017, 2021 and 2023, and recovers the deficit in the minority of years when the benchmark suffers. The three years after the holdout boundary follow the same pattern: drawdown wins in all three, Sharpe wins in two.

Two years merit comment. In 2021 the framework loses on both measures, the only such year, because the mixture model spent much of that year in the transitional state during what turned out to be a sustained advance. In 2018 the framework loses on Sharpe despite a smaller loss in absolute terms, an arithmetic consequence of a smaller negative return divided by a much smaller volatility.

Episodic Rather Than Persistent:

The rolling window analysis bears on the question of performance persistence, and answers it against the framework. The strategy's risk-adjusted return profile does not consistently surpass that of a passive benchmark; it surpasses it in slightly more windows than not, and the excess returns are clustered in the specific conditions catalogued in Table 6. The advantage is therefore episodic rather than persistent, which is what Table 10 leads one

to expect and what the stress-episode results describe from the other direction. A reader evaluating this framework should expect long stretches of underperformance punctuated by periods in which the accumulated deficit is recovered, and should size the position accordingly.

4.3 Robustness and Attribution

The subsections that follow separate the framework into its components, test it outside the sample on which it was specified, and report the inference that bounds every claim made above.

4.3.1 Out-of-Sample Holdout, 2024–2026

Every specification decision in this study—the universe, the feature set, the number of regimes, the risk budget, the rebalancing interval and the cost assumption—was fixed before any session after December 2023 was examined. The 665 sessions from January 2024 to August 2026 therefore function as a holdout rather than as a further robustness check on the same data. Table 11 reports both blocks side by side.

Table 11 Performance before and after the holdout boundary. The two blocks are disjoint; the full-sample results of Table 5 pool them.

	2008–2023 ($n = 4,024$)		2024–2026 ($n = 665$)	
	Sharpe	Max DD	Sharpe	Max DD
Regime-conditional strategy	0.738	−18.98%	1.614	−5.48%
Equal-weight benchmark	0.532	−36.08%	1.391	−10.13%
Inverse-volatility benchmark	0.764	−24.09%	1.397	−6.70%
Equal-weight at strategy volatility	0.560	−17.66%	1.378	−7.04%
No regime timing, full exposure	0.785	−22.53%	1.710	−6.47%

In the holdout the framework records a higher Sharpe ratio than all three benchmarks, 1.614 against 1.391, 1.397 and 1.378, and a smaller maximum drawdown than all three, −5.48 per cent against −10.13, −6.70 and −7.04 per cent. Two of those comparisons reverse the in-sample ordering: the inverse-volatility benchmark and the volatility-matched equal-weighted benchmark both beat the framework on Sharpe ratio before 2024 and both trail it afterwards. The deepest equal-weighted drawdown in the holdout, the 34-session decline from 20 February to 8 April 2025, returned −10.05% for the benchmark against −4.63% for the framework, an advantage of 5.42 percentage points that is consistent with the four earlier episodes in Table 6.

Three qualifications limit what this establishes. First, 665 sessions are too few to settle anything on their own: the bootstrap interval for the Sharpe difference in the holdout is $[-0.352, 0.742]$, wider than the full-sample interval and comfortably containing zero, and the framework again earns less in absolute terms, giving up 3.07 percentage points of annual return ($p = 0.348$). Second, the holdout contains no episode comparable to 2008 or 2020; every series in it posts a Sharpe ratio above 1.3, so the period is benign and the relevant information is the ordering rather than the level. Third, the ablation ordering is unchanged—the variant with no regime timing at full exposure again records the highest Sharpe ratio, 1.710—so the holdout does not rescue any claim that Section 4.3.4 rejects.

What the holdout does establish is narrower and still worth stating. The framework’s defensive behaviour is not an artefact of the sample on which it was specified. It was built

without reference to these sessions, and on them it continues to reduce drawdown against every benchmark considered while remaining competitive on risk-adjusted return.

4.3.2 Robustness Across Specifications

The primary configuration fixes a random seed, a covariance estimation window and a rebalancing interval. Table 12 reports every combination of three seeds, two estimation windows and three rebalancing intervals, so that the reported result can be located within the distribution of specifications rather than presented in isolation.

Table 12 Specification sweep. All eighteen cells reported; the last two columns record whether the cell beats the equal-weight benchmark on each measure. Sharpe ratios here are the annualised arithmetic mean over volatility, so that all eighteen cells are computed identically; on the same basis the equal-weight benchmark attains 0.670 with a maximum drawdown of -36.1% .

Seed	Cov. window	Rebalance	Sharpe	Max DD	Sharpe win	DD win
0	126	10	0.868	-20.7%	yes	yes
0	126	21	0.954	-19.6%	yes	yes
0	126	42	0.924	-19.7%	yes	yes
0	252	10	0.823	-20.8%	yes	yes
0	252	21	0.905	-19.0%	yes	yes
0	252	42	0.899	-20.4%	yes	yes
7	126	10	0.881	-20.6%	yes	yes
7	126	21	0.956	-19.6%	yes	yes
7	126	42	0.912	-19.6%	yes	yes
7	252	10	0.818	-21.1%	yes	yes
7	252	21	0.891	-19.0%	yes	yes
7	252	42	0.888	-20.4%	yes	yes
42	126	10	0.853	-20.6%	yes	yes
42	126	21	0.928	-19.6%	yes	yes
42	126	42	0.893	-19.7%	yes	yes
42	252	10	0.813	-21.1%	yes	yes
42	252	21	0.887	-19.0%	yes	yes
42	252	42	0.853	-20.4%	yes	yes

Every one of the eighteen specifications beats the equal-weighted benchmark on both the Sharpe ratio and maximum drawdown. The Sharpe ratio ranges from 0.813 to 0.956 and the maximum drawdown from -21.1 to -19.0 per cent. The primary configuration appears in this table at 0.887, which is the median of the eighteen values: it is not the best cell, and the best cell exceeds it by 0.069. Sensitivity to the random seed is small, which indicates that the mixture fit is stable; sensitivity to the rebalancing interval is larger, with ten-session rebalancing consistently weakest because it incurs more turnover for no additional information.

The same sweep is unfavourable on one comparison. In none of the eighteen cells does the framework beat the volatility-matched equal-weighted benchmark on maximum drawdown. This is a consistent finding rather than an artefact of the primary configuration, and it bounds the drawdown claim precisely: the framework reduces drawdown relative to an unlevered diversified benchmark, not relative to every portfolio held at the same volatility.

4.3.3 External Validation of the Sentiment Composite

Naming the second composite a sentiment measure is a claim about what it measures, and a claim of that kind is testable without reference to whether it makes money. This subsection tests it; Section 4.2.5 asks the separate question of whether it forecasts anything.

The composite is aggregated to the market level by averaging each of its four inputs across the ten instruments, standardising each average over the sample and combining the four with the signs of Table 4. The resulting series is compared with four published measures, none of which is observed anywhere in the framework: the University of Michigan survey of consumer sentiment, the Cboe volatility index, the St. Louis Fed financial stress index and the Chicago Fed national financial conditions index. The first is a household survey and contains no market data at all, which makes it the most demanding of the four. A sentiment measure should move with it and against the remaining three.

Table 13 External validation of the sentiment composite, January 2008 to August 2026.

Newey–West *t*-statistics in parentheses, with twenty-one lags for daily, eight for weekly and six for monthly data. The change columns correlate first differences and so are not driven by any common trend in levels.

Series	Expected	Levels	Changes
Michigan consumer sentiment, monthly (<i>n</i> = 222)	+	+0.262 (+1.91)	+0.240 (+3.54)
Cboe volatility index, daily (<i>n</i> = 4,689)	–	–0.541 (–3.62)	–0.112 (–3.02)
St. Louis Fed financial stress index, weekly (<i>n</i> = 972)	–	–0.606 (–2.71)	–0.254 (–3.68)
Chicago Fed financial conditions index, weekly (<i>n</i> = 972)	–	–0.513 (–2.46)	–0.123 (–2.18)

Every correlation in Table 13 carries the expected sign, and every one of the four is significant in first differences at the five per cent level. The survey measure is the informative case. A correlation of +0.240 between monthly changes in a composite built from exchange turnover and monthly changes in what households tell a telephone interviewer is not a mechanical artefact of shared construction, because the two series share no inputs; it is evidence that the composite is picking up the quantity its name asserts. The financial stress and financial conditions indices, which do contain market data, correlate more strongly in levels at –0.606 and –0.513.

The relationship is stable in sign though not in magnitude. Splitting the sample into four disjoint blocks—2008–2013, 2014–2019, 2020–2023 and 2024–2026—the sign is correct in all sixteen series-by-block combinations. Magnitudes are largest in the two blocks containing crises, reaching –0.823 against the stress index over 2008–2013, and smallest over the calm 2014–2019 block, where the correlation with that index falls to –0.196. This is the expected pattern rather than a defect: a measure of stress has little to discriminate when there is little stress. Replacing the full-sample standardisation with an expanding-window standardisation that uses only past data leaves every sign unchanged, at +0.182, –0.341, –0.406 and –0.319 respectively.

The second test bears directly on the framework’s central mechanism. If the identified market states are states of sentiment and volatility, the composite should differ across them. Table 14 evaluates the composite at the 223 rebalance dates for which a regime label exists.

The ordering is monotone and steep, and the differences are unlikely to be accidental: one-way analysis of variance gives $F = 19.34$ with $p = 1.8 \times 10^{-8}$, and the rank-based

Table 14 Sentiment composite by identified market state, measured at the rebalance dates. The states are labelled by fitted volatility, not by sentiment, and the sentiment composite plays no part in fitting them.

Identified state	<i>n</i>	Mean	SD
Calm	160	+0.073	0.407
Transitional	56	−0.140	0.382
Stress	7	−0.831	0.735

Kruskal–Wallis test, which does not assume normality and is the safer of the two given seven observations in the stress state, gives $H = 29.54$ with $p = 3.9 \times 10^{-7}$. The mixture model is fitted on realised volatility and short-horizon momentum; the sentiment composite is not among its inputs. That the states separate this sharply on a variable withheld from their construction is what justifies describing them as sentiment and volatility states rather than volatility states alone.

Two limits should be read alongside these results. The stress state contains seven rebalance dates, so its mean is estimated imprecisely even though the test statistic is large. And nothing established here concerns forecasting. A composite can measure sentiment accurately and still fail to predict returns, which is precisely what Section 4.2.5 documents; validity and profitability are separate properties, and conflating them is a common error in this literature.

4.3.4 Component Ablation

The framework has two active layers, signal fusion and the regime risk budget. Table 15 removes each in turn. The comparison requires care, because scaling a portfolio toward cash changes its return and its volatility together; comparing a de-risked strategy against a fully invested variant therefore confounds the timing of risk with its level. The first ablation holds average invested exposure fixed at the framework’s own 92.1 per cent and applies it constantly, so that only the timing differs.

Table 15 Component ablation. The matched-exposure control holds the same average invested exposure as the full framework but applies it without regard to the regime.

Configuration	CAGR	Sharpe	Max DD	Exposure	<i>p</i>
Full framework	5.65%	0.878	−18.98%	92.1%	—
No regime timing, same average exposure	5.43%	0.795	−22.26%	92.1%	0.695
No regime timing, full exposure	6.91%	0.923	−22.53%	100.0%	0.016
No signal layer	5.75%	0.908	−18.87%	92.1%	0.650
Equal-weight benchmark	8.45%	0.628	−36.08%	100.0%	—

The first comparison is the one the framework is built on, and it succeeds. Holding average exposure fixed, allocating that exposure according to the regime rather than uniformly raises the Sharpe ratio from 0.795 to 0.878 and improves maximum drawdown from −22.26 to −18.98 per cent. The difference in mean return is not significant ($p = 0.695$), which is the expected signature of a pure risk-timing effect: the same average risk is carried, but it is carried at better moments. This is the only comparison in the study in which the framework improves both measures at once, and it is the claim the paper rests on.

The second comparison bounds the cost of that timing. Against a fully invested variant, the framework gives up approximately 1.3 percentage points of annual return, and the difference is statistically significant ($p = 0.016$). De-risking is not free, and its Sharpe

ratio under this comparison is lower, 0.878 against 0.923. What the framework buys with that return is 3.55 percentage points of maximum drawdown. Whether that exchange is worthwhile is a preference, not a fact, and the paper reports both sides of it.

The third comparison is a null, and it concerns the layer the framework was named for. Removing the signal fusion layer entirely—setting expected returns to zero so that the optimiser becomes a constrained minimum-variance problem—changes the Sharpe ratio from 0.878 to 0.908 and the maximum drawdown from -18.98 to -18.87 per cent. Both movements favour removal, and neither is statistically significant ($p = 0.650$). The signal fusion layer contributes nothing detectable to portfolio outcomes in this universe, and its point estimate is mildly adverse. Section 4.2.5 establishes why.

Form of the Risk Budget:

Table 16 varies the risk budget itself, including the absolute-target formulation used in the framework as originally specified.

Table 16 Risk budget specifications. Relative budgets are stated as the fraction of the sleeve's own volatility retained in the calm, transitional and stress states.

Budget	CAGR	Sharpe	Max DD	Exposure
Relative 1.00 / 0.75 / 0.50 (primary)	5.65%	0.878	-18.98%	92.1%
Relative 1.00 / 0.80 / 0.60	5.77%	0.871	-19.95%	93.7%
Relative 1.00 / 0.60 / 0.30	5.31%	0.881	-16.62%	87.6%
Relative 1.00 / 0.50 / 0.25	5.21%	0.891	-15.71%	84.9%
Absolute targets, 20 / 15 / 10% annualised	6.93%	0.923	-22.86%	100.0%

Two findings follow. First, the Sharpe ratio varies little across the four relative budgets, from 0.871 to 0.891, while maximum drawdown improves monotonically as the budget tightens, from -19.95 to -15.71 per cent. The result is therefore not an artefact of the particular multipliers chosen; the mildest of the four is reported as primary precisely because it is the least aggressive, and the tighter budgets would have produced a marginally better headline on both measures.

Second, the absolute-target formulation is inert. Its mean invested exposure is 99.99 per cent, and the constraint binds in one rebalance out of 224. Targets of 10 to 20 per cent annualised volatility are never reached by a sleeve whose own realised volatility is 6.4 per cent, so under that formulation the regime layer is present in the specification but has no effect on the portfolio. Its apparently superior Sharpe ratio of 0.923 is simply the fully invested portfolio of Table 15, reached by a different route. This is the practical reason the relative formulation is adopted: an absolute volatility target is not scale-free, and a target calibrated for one universe silently disables the mechanism in another.

4.3.5 Statistical Significance

Table 17 reports inference on the daily excess return series and on the Sharpe difference.

The results must be stated carefully. The framework's mean return is below the benchmark's, and that shortfall is not statistically significant ($p = 0.171$). The Sharpe difference of $+0.250$ has a bootstrap median of $+0.242$ and a 95 per cent interval of $[-0.102, 0.574]$ that includes zero; the bootstrap probability that the difference is not positive is 0.088. On an eighteen-year sample, therefore, the Sharpe improvement is not significant at conventional levels, and the paper does not claim that it is. Extending the sample by the 665 sessions of the holdout narrowed that interval—its lower bound moves

Table 17 Inference against the equal-weighted benchmark. Bootstrap intervals are from a stationary block bootstrap with 21-session blocks and 2,000 replications.

Mean excess return over benchmark, annualised	−3.31%
Paired t -statistic	−1.369
p -value	0.171
Newey–West t -statistic, ten lags	−1.699
Strategy Sharpe ratio, 95% bootstrap interval	[0.394, 1.368]
Benchmark Sharpe ratio, 95% bootstrap interval	[0.180, 1.165]
Sharpe difference, median	+0.242
Sharpe difference, 95% bootstrap interval	[−0.102, 0.574]
Bootstrap probability that the difference is not positive	0.088

from −0.194 on the pre-2024 data to −0.102—without crossing zero, which is the expected behaviour of a real but modest effect measured on a sample that remains too short.

What is robust is not the point estimate but its consistency. The Sharpe advantage appears in 18 of 18 specifications, the drawdown advantage in 18 of 18 and in 18 of 19 calendar years, and the outperformance in all four stress episodes. Eighteen years of daily data contain few independent observations of the events that matter for a drawdown claim—four, on the evidence of Table 6—and no amount of daily sampling converts that into statistical power (Harvey and Liu, 2015). The appropriate conclusion is that the drawdown reduction is consistent and mechanically explicable, and that the Sharpe improvement, while positive under every specification examined, is not established at conventional significance.

4.3.6 Secondary Universe: Mega-Capitalisation Equities

The framework was also run without modification on the ten mega-capitalisation US equities of the original study, over the same 2018–2026 download window, giving 1,920 evaluated sessions from January 2019. Table 18 reports the outcome, which is unfavourable on risk-adjusted return.

Table 18 Secondary universe: ten mega-capitalisation US equities, 2019–2026

	CAGR	Volatility	Sharpe	Max DD
Regime-conditional strategy	17.02%	14.87%	1.145	−20.45%
Equal-weight benchmark	33.52%	24.45%	1.371	−33.63%
Equal-weight at strategy volatility	20.08%	14.87%	1.350	−21.41%
No regime timing, same average exposure	18.82%	16.82%	1.119	−30.36%

The framework loses on risk-adjusted return, 1.145 against 1.371, and gives up 15.09 percentage points of annual return, a shortfall that is statistically significant ($t = -2.791$, $p = 0.005$). Its defensive behaviour nevertheless survives intact: maximum drawdown improves from −33.63 to −20.45 per cent, it wins on drawdown in 8 of 8 calendar years while winning on Sharpe ratio in only 3, and it outperforms in both stress episodes this window contains, by 12.50 percentage points during the 2020 crash and 12.60 percentage points during the 2022 bear market.

The regime layer itself continues to behave as the primary results describe. Against the matched-exposure control it improves the Sharpe ratio slightly, 1.145 against 1.119, and improves maximum drawdown substantially, from −30.36 to −20.45 per cent. The risk-timing effect identified in Section 4.3.4 therefore reproduces in a universe on which the framework loses overall, which is the relevant test of whether that effect is specific to the

primary sample. What fails here is not the regime layer but the decision to hold the resulting portfolio in place of the benchmark.

The reason is a property of the universe rather than of the framework. Ten mega-capitalisation US equities are close to a single factor bet: correlations are high, there is no low-volatility asset to rotate into, and the risky sleeve therefore carries roughly 15 per cent annualised volatility against 6 per cent in the multi-asset case. De-risking in such a universe can only move capital to cash, so the framework systematically forgoes return during an exceptional advance and recovers only part of it in drawdowns. This bounds the applicability of the framework directly: it requires a universe with genuine dispersion in asset volatility, and it should not be expected to add value in a concentrated, single-sector one. The originally submitted version of this study used exactly such a universe, which is why its results are superseded here.

4.4 Theoretical and Practical Implications

4.4.1 Methodological Contributions

- **Signal-allocation bridge.** The research demonstrates a systematic framework for translating predictive signals into executable portfolio decisions, addressing a critical gap in the literature. What it additionally shows is that the bridge can be built and the signal crossing it can still be uninformative, which is why each layer is tested separately below.
- **Regime adaptation.** The regime-dependent risk budget improves both risk-adjusted return and drawdown at matched average exposure. The regime-dependent *weighting* mechanism, by contrast, is not supported: the conditional efficacy it presumes is not present in the data.
- **Practical implementation.** The incorporation of transaction costs and realistic constraints ensures the approach is implementable in live trading environments, and the cost assumption is shown to act as a regulariser rather than merely as a deduction.

The framework's contribution is therefore narrower than the one claimed for it at the outset, and better supported. Three findings survive the evaluation above. First, splitting the problem into relative composition and total risk, rather than letting one constraint set determine both, makes each part independently testable, and is what allows the ablation in Table 15 to be constructed at all. Second, regime conditioning earns its place through timing rather than level: at identical average exposure it improves the Sharpe ratio from 0.795 to 0.878 and maximum drawdown from -22.26 to -18.98 per cent with no significant difference in mean return ($p = 0.695$), at a measured cost of approximately 1.3 percentage points of annual return against a fully invested variant, significantly so, for 3.6 percentage points of maximum drawdown. Third, the form of the risk budget matters more than its calibration—the four relative budgets in Table 16 produce Sharpe ratios within 0.020 of one another, while the absolute-target formulation, identical in intent, is inert, binding in one rebalance out of 224. An absolute volatility target is not scale-free, so a specification moved between universes without re-calibration can silently disable the mechanism it was written to implement.

What the Results Do Not Establish:

Two claims the framework was designed to support are not supported. The signal fusion layer does not contribute: removing it improves the Sharpe ratio from 0.878 to 0.908 and

leaves maximum drawdown unchanged, neither movement significant, and Table 9 explains why—both composites are significantly negatively related to next-session cross-sectional returns over the full sample. The regime-conditional weighting schedule has no measured basis either. It presumes that sentiment information becomes relatively more informative under stress; the point estimates run the other way and the calm-to-stress contrast is not significant ($t = -1.46$, $p = 0.145$). Split at the holdout boundary, the fused coefficient is -0.039 before 2024 and $+0.009$ after, so even the negative relationship is era-specific and could not have been exploited by inverting the composites. Nor is the improvement in risk-adjusted return statistically significant: the Sharpe difference of $+0.250$ carries a bootstrap interval of $[-0.102, 0.574]$, and adding the 665 sessions of the holdout narrowed that interval without moving it clear of zero. Consistency across 18 of 18 specifications, 18 of 19 calendar years and 4 of 4 episodes is evidence of robustness rather than of significance. The two should not be conflated, and this paper does not conflate them.

Implications for Practice:

A regime-conditional risk budget is a reasonable component of a multi-asset allocation process whose objective includes drawdown control and which can sacrifice some return for it. It should be implemented in relative rather than absolute form, evaluated against a volatility-matched benchmark rather than an unlevered one, and not expected to help in a universe without genuine dispersion in asset volatility. The results also caution against a common design pattern: the framework as originally specified contained two layers whose contribution had never been isolated, one inert by construction and the other pointed in the wrong direction, and neither defect was visible in aggregate performance (López de Prado, 2016).

4.4.2 Limitations and Future Research

Several limitations bound what can be concluded.

- The analysis relies on simplified proxy variables for market sentiment, foregoing the capabilities of contemporary large language model processing of textual data.
- The weighting parameters applied within the regime-dependent fusion mechanism are statically defined, lacking an adaptive framework for their continuous optimisation.
- The empirical validation is confined to a constrained asset universe and a specific historical timeframe, potentially limiting the generalisability of the findings.

Each bears on the results in a specific way. The universe is small: ten exchange-traded funds demonstrate the mechanism but do not establish that it generalises to larger or differently composed universes, and the 25 per cent position cap binds on average, so the optimiser's freedom is partly set by the constraint rather than by the risk model. The evaluation is also a single historical path—eighteen years is long by the standards of this literature and short for inference about tail events, and the four stress episodes are not independent draws from a stationary process.

The signals are market-derived proxies rather than textual sentiment: no news, filings or social media text is used at any point, as stated in Section 3.1.3. The null reported in Section 4.2.5 is therefore a null about these proxies, not evidence that textual sentiment would fail in the same role; establishing that would require a time-stamped news corpus covering the traded instruments, which for broad exchange-traded funds does not exist at the instrument level. The regime model is deliberately simple—a three-component Gaussian

mixture on two features is interpretable and stable, but imposes conditional normality within states, treats observations as conditionally independent given the state, and, as Table 7 shows, is not the specification either information criterion would select. Finally, the risk-free rate is taken as zero and cash assumed to earn nothing; over a sample spanning both the near-zero rates of 2009–2015 and the higher rates of 2023 this understates the return on the framework’s cash holding and so understates its performance against the fully invested benchmark. That bias runs against the framework, which is the conservative choice, but it is a simplification.

Research Extensions:

Four extensions follow directly from the limitations above.

- **Advanced sentiment integration.** A logical extension involves the incorporation of language-model-driven sentiment analysis to derive more sophisticated indicators of market affect, applied to a universe where an instrument-level news corpus exists. This would convert the proxy null of Section 4.2.5 into a genuine test of the fusion hypothesis.
- **Dynamic optimisation frameworks.** The implementation of reinforcement learning algorithms could facilitate the dynamic calibration of regime-specific parameters, moving beyond the current static configuration. This is subject to the overfitting concerns documented by Harvey and Liu (2015) and López de Prado (2016).
- **Expanded portfolio scope.** The framework’s robustness could be further tested through application to global, multi-asset portfolios encompassing equities, fixed income, commodities and currencies. Such a test would establish whether the volatility-dispersion condition identified in Section 4.3.6 is the binding requirement it appears to be.
- **Alternative data synthesis.** Future work could enhance the feature set by integrating a broader spectrum of alternative data, such as supply chain information or electronic market microstructure data, which would widen the test of the fusion hypothesis beyond the price and volume aggregates used here.
- **Alternative regime formulations.** Replacing the mixture model with a formulation that represents state persistence explicitly, such as a hidden Markov or jump model (Li et al., 2025; Luo and Mulvey, 2026), would remove the conditional-independence assumption noted above.

5 Conclusion

The empirical findings of this study affirm one half of the central thesis that a regime-sensitive framework for signal integration and portfolio construction delivers an improved risk-return dynamic compared to static approaches, and refute the other. Superior capital retention during market downturns, a lower volatility signature in bullish phases, and a rolling Sharpe ratio positive in the majority of windows demonstrate the efficacy of one of the model’s key components—its ability to adjust total risk to evolving market conditions—together with an implementation strategy that accounts for transaction costs. They do not support its probabilistic fusion of diverse signal types. This paper set out to close the step between a predictive signal and an executable portfolio, and to determine which parts of a

regime-conditional framework are responsible for its behaviour. Both objectives are met, though not in the direction originally anticipated.

Evaluated on ten liquid multi-asset instruments over eighteen years under a walk-forward protocol with transaction costs, the framework raises the Sharpe ratio from 0.628 to 0.878 and reduces maximum drawdown from -36.1 to -19.0 per cent against an equal-weighted benchmark, while earning a lower compound return. The advantage holds in all four stress episodes, in 18 of 19 calendar years on drawdown, in all eighteen specifications examined, and in a held-out final block that postdates every specification decision—but it is consistent rather than statistically significant, its bootstrap interval including zero. Ablation locates its source: at identical average exposure, timing risk by the identified market state improves both measures with no difference in mean return, and the form of the risk budget proves more consequential than its calibration, since the absolute targets of the original specification bind in one rebalance out of 224 and leave the layer inert.

The signal fusion layer, which gave the framework its name, does not contribute. Removing it improves the Sharpe ratio, the information coefficients of both composites are significantly negative over the full sample, and the regime-conditional weighting schedule has no measured basis, since the calm-to-stress contrast it presumes is not significant and its point estimates run the other way. Nor is the negative relationship stable enough to invert, since it disappears in the holdout. This null is reported in full, with the coefficients that produce it, because a component-level negative result is more useful than an aggregate number that conceals which parts of an architecture are doing the work.

The study's contribution is therefore a specification and a method of testing it: a two-step separation of composition from total risk that makes each layer independently falsifiable, an ablation design that isolates risk timing from risk level by matching average exposure, and a documented account of two failure modes—an inert absolute risk budget, and a signal-to-allocation conversion in which mismatched units rather than information determine the solution. The framework is useful within stated bounds: it requires a universe with genuine dispersion in asset volatility, trades return for drawdown at a measured rate, and should be evaluated against a volatility-matched rather than an unlevered benchmark. Outside those bounds, as the mega-capitalisation results show, it does not add value. Performance of this kind across varied and often challenging market scenarios highlights the necessity of addressing the theoretical disconnect between signal generation and allocation, and the study offers both a methodological foundation for a new generation of adaptive investment models and an empirical account of what such models can and cannot be shown to achieve—provided each layer of the resulting architecture is tested separately rather than credited with the performance of the whole.

Acknowledgements

Acknowledgements are withheld to preserve anonymity during review.

Author contributions

Anandkumar Pardeshi: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft.

Sujata Deshmukh: Supervision, Validation, Writing – review & editing, Research coordination.

Conflict of interest

The authors declare no conflict of interest.

Data and code availability

All market data used in this study are publicly available daily price and volume series obtained from Yahoo Finance for the instruments listed in Section 3.1. The analysis code implementing the framework, the specification sweep and the ablations is available from the corresponding author on reasonable request.

AI tool usage declaration

During the preparation of this work, the authors used Claude (Anthropic) and ChatGPT (OpenAI) in order to improve the readability and language of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No part of the analysis, the generation of results, or the interpretation of findings was delegated to such tools.

References

- Alsheebah, F.M.M. and Al-Fuhaidi, B.A. (2024) 'Emerging Stock Market Prediction Using GRU Algorithm: Incorporating Endogenous and Exogenous Variables', *IEEE Access*, Vol. 12, pp. 132964–132971, doi: 10.1109/ACCESS.2024.3444699.
- Anbaee Farimani, S., Jahan, M.V. and Milani Fard, A. (2024) 'An Adaptive Multimodal Learning Model for Financial Market Price Prediction', *IEEE Access*, Vol. 12, pp. 121846–121863, doi: 10.1109/ACCESS.2024.3441029.
- Asimit, V., Chong, W.F., Tunaru, R. and Zhou, F. (2025) 'Portfolio selection and risk sharing via risk budgeting', *Insurance: Mathematics and Economics*, Vol. 125, pp. 103139, doi: 10.1016/j.insmatheco.2025.103139.
- Bacco, L., Petrosino, L., Arganese, D., Vollero, L., Papi, M. and Merone, M. (2024) 'Investigating Stock Prediction Using LSTM Networks and Sentiment Analysis of Tweets Under High Uncertainty: A Case Study of North American and European Banks', *IEEE Access*, Vol. 12, pp. 122239–122248, doi: 10.1109/ACCESS.2024.3450311.
- Baker, M. and Wurgler, J. (2007) 'Investor Sentiment in the Stock Market', *Journal of Economic Perspectives*, Vol. 21, No. 2, pp. 129–151, doi: 10.1257/jep.21.2.129.
- Chou, J.-S. and Nguyen, T.-K. (2018) 'Forward Forecast of Stock Price Using Sliding-Window Metaheuristic-Optimized Machine-Learning Regression', *IEEE Transactions on Industrial Informatics*, Vol. 14, No. 7, pp. 3132–3142, doi: 10.1109/TII.2018.2794389.

- DeMiguel, V., Garlappi, L. and Uppal, R. (2009) 'Optimal Versus Naive Diversification: How Inefficient is the $1/N$ Portfolio Strategy?', *Review of Financial Studies*, Vol. 22, No. 5, pp. 1915–1953, doi: 10.1093/rfs/hhm075.
- George, M. and Murugesan, R. (2025) 'Sentiment analysis on stocks: a hybrid feature extraction technique on 14 classifiers', *International Journal of Applied Decision Sciences*, Vol. 18, No. 1, pp. 84–112, doi: 10.1504/IJADS.2025.143083.
- Grinold, R.C. and Kahn, R.N. (2000) *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*, 2nd ed., McGraw-Hill, New York, NY.
- Ha, Y.S., Kang, J., Kim, J. and Chun, D. (2026) 'Machine learning-based portfolio optimization: comparative analysis with the all-weather portfolio strategy', *Financial Innovation*, Vol. 12, No. 1, doi: 10.1186/s40854-026-00927-8.
- Hamilton, J.D. (1989) 'A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle', *Econometrica*, Vol. 57, No. 2, pp. 357–384, doi: 10.2307/1912559.
- Haryono, A.T., Sarno, R. and Sungkono, K.R. (2023) 'Transformer-Gated Recurrent Unit Method for Predicting Stock Price Based on News Sentiments and Technical Indicators', *IEEE Access*, Vol. 11, pp. 77132–77146, doi: 10.1109/ACCESS.2023.3298445.
- Haryono, A.T., Sarno, R., Anggraini, R.N.E. and Sungkono, K.R. (2024) 'Permuted Temporal Kolmogorov-Arnold Networks for Stock Price Forecasting Using Generative Aspect-Based Sentiment Analysis', *IEEE Access*, Vol. 12, pp. 178672–178689, doi: 10.1109/ACCESS.2024.3506658.
- Harvey, C.R. and Liu, Y. (2015) 'Backtesting', *The Journal of Portfolio Management*, Vol. 42, No. 1, pp. 13–28, doi: 10.3905/jpm.2015.42.1.013.
- Hood, B.T. and Raughtigan, C. (2025) 'Volatility Targeting Is Trendy: How Trend Following Explains Alpha in Volatility-Managed Strategies', *The Journal of Portfolio Management*, Vol. 52, No. 1, pp. 100–121, doi: 10.3905/jpm.2025.1.764.
- Hu, X. and Yang, J. (2024) 'Group Penalized Multinomial Logit Models and Stock Return Direction Prediction', *IEEE Transactions on Information Theory*, Vol. 70, No. 6, pp. 4297–4318, doi: 10.1109/TIT.2024.3376751.
- Huang, Q., Yang, J., Feng, X., Liew, A.W.-C. and Li, X. (2020) 'Automated Trading Point Forecasting Based on Bicluster Mining and Fuzzy Inference', *IEEE Transactions on Fuzzy Systems*, Vol. 28, No. 2, pp. 259–272, doi: 10.1109/TFUZZ.2019.2904920.
- Jung, H.D. and Jang, B. (2024) 'Enhancing Financial Sentiment Analysis Ability of Language Model via Targeted Numerical Change-Related Masking', *IEEE Access*, Vol. 12, pp. 50809–50820, doi: 10.1109/ACCESS.2024.3385855.
- Kabbani, T. and Duman, E. (2022) 'Deep Reinforcement Learning Approach for Trading Automation in the Stock Market', *IEEE Access*, Vol. 10, pp. 93564–93574, doi: 10.1109/ACCESS.2022.3203697.

- Khodamoradi, T., Salahi, M. and Najafi, A.R. (2023) 'Multi-intervals robust mean-conditional value-at-risk portfolio optimisation with conditional scenario reduction technique', *International Journal of Applied Decision Sciences*, Vol. 16, No. 2, pp. 237–256, doi: 10.1504/IJADS.2023.129475.
- Khuwaja, P., Khowaja, S.A. and Dev, K. (2023) 'Adversarial Learning Networks for FinTech Applications Using Heterogeneous Data Sources', *IEEE Internet of Things Journal*, Vol. 10, No. 3, pp. 2194–2201, doi: 10.1109/JIOT.2021.3100742.
- Kritzman, M., Page, S. and Turkington, D. (2012) 'Regime Shifts: Implications for Dynamic Strategies', *Financial Analysts Journal*, Vol. 68, No. 3, pp. 22–39, doi: 10.2469/faj.v68.n3.3.
- Ledoit, O. and Wolf, M. (2004) 'A well-conditioned estimator for large-dimensional covariance matrices', *Journal of Multivariate Analysis*, Vol. 88, No. 2, pp. 365–411, doi: 10.1016/S0047-259X(03)00096-4.
- Lee, R.S.T. (2004) 'iJADE Stock Advisor: An Intelligent Agent Based Stock Prediction System Using Hybrid RBF Recurrent Network', *IEEE Transactions on Systems, Man, and Cybernetics, Part A: Systems and Humans*, Vol. 34, No. 3, pp. 421–428, doi: 10.1109/TSMCA.2004.824871.
- Lee, J.W., Park, J., O, J., Lee, J. and Hong, E. (2007) 'A Multiagent Approach to Q -Learning for Daily Stock Trading', *IEEE Transactions on Systems, Man, and Cybernetics, Part A: Systems and Humans*, Vol. 37, No. 6, pp. 864–877, doi: 10.1109/TSMCA.2007.904825.
- Li, A.W. and Bastos, G.S. (2020) 'Stock Market Forecasting Using Deep Learning and Technical Analysis: A Systematic Review', *IEEE Access*, Vol. 8, pp. 185232–185242, doi: 10.1109/ACCESS.2020.3030226.
- Li, X., Chen, J., Tao, X. and Ji, Y. (2025) 'Regime-Switching Asset Allocation Using a Framework Combining a Jump Model and Model Predictive Control', *Mathematics*, Vol. 13, No. 17, pp. 2837, doi: 10.3390/math13172837.
- Liu, G. and Wang, X. (2019) 'A Numerical-Based Attention Method for Stock Market Prediction With Dual Information', *IEEE Access*, Vol. 7, pp. 7357–7367, doi: 10.1109/ACCESS.2018.2886367.
- Liu, C., Arulappan, A., Naha, R., Mahanti, A., Kamruzzaman, J. and Ra, I.-H. (2024) 'Large Language Models and Sentiment Analysis in Financial Markets: A Review, Datasets, and Case Study', *IEEE Access*, Vol. 12, pp. 134041–134061, doi: 10.1109/ACCESS.2024.3445413.
- López de Prado, M. (2016) 'Building Diversified Portfolios that Outperform Out of Sample', *The Journal of Portfolio Management*, Vol. 42, No. 4, pp. 59–69, doi: 10.3905/jpm.2016.42.4.059.
- Loughran, T. and McDonald, B. (2011) 'When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks', *The Journal of Finance*, Vol. 66, No. 1, pp. 35–65, doi: 10.1111/j.1540-6261.2010.01625.x.

- Lu, X., Qiu, J., Yang, Y., Zhang, C., Lin, J. and An, S. (2025) 'Large Language Model-Based Bidding Behavior Agent and Market Sentiment Agent-Assisted Electricity Price Prediction', *IEEE Transactions on Energy Markets, Policy and Regulation*, Vol. 3, No. 2, pp. 223–235, doi: 10.1109/TEMPR.2024.3518624.
- Luo, Y. and Mulvey, J.M. (2026) 'Regime-Aware Asset Allocation with Dual-Regime Signals and Regime-Dependent Asset Selection', *The Journal of Financial Data Science*, Vol. 8, No. 3, pp. 104–134, doi: 10.3905/jfds.2026.013.
- Markowitz, H. (1952) 'Portfolio Selection', *The Journal of Finance*, Vol. 7, No. 1, pp. 77–91, doi: 10.2307/2975974.
- Moodi, F., Rafsanjani, A.J., Zarifzadeh, S. and Chahooki, M.A.Z. (2024) 'Fusion of Technical Indicators and Sentiment Analysis in a Hybrid Framework of Deep Learning Models for Stock Price Movement Prediction', *IEEE Access*, Vol. 12, pp. 195696–195709, doi: 10.1109/ACCESS.2024.3519556.
- Moreira, A. and Muir, T. (2017) 'Volatility-Managed Portfolios', *The Journal of Finance*, Vol. 72, No. 4, pp. 1611–1644, doi: 10.1111/jofi.12513.
- Mu, G., Gao, N., Wang, Y. and Dai, L. (2023) 'A Stock Price Prediction Model Based on Investor Sentiment and Optimized Deep Learning', *IEEE Access*, Vol. 11, pp. 51353–51367, doi: 10.1109/ACCESS.2023.3278790.
- Nayak, S.C., Dehuri, S. and Cho, S.-B. (2022) 'Intelligent Financial Forecasting With an Improved Chemical Reaction Optimization Algorithm Based Dendritic Neuron Model', *IEEE Access*, Vol. 10, pp. 130921–130943, doi: 10.1109/ACCESS.2022.3229314.
- Newey, W.K. and West, K.D. (1987) 'A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix', *Econometrica*, Vol. 55, No. 3, pp. 703–708, doi: 10.2307/1913610.
- Ónozó, L.R., Arthur, F.V. and Gyires-Tóth, B. (2024) 'Leveraging LLMs for Financial News Analysis and Macroeconomic Indicator Nowcasting', *IEEE Access*, Vol. 12, pp. 160529–160547, doi: 10.1109/ACCESS.2024.3488363.
- Otabek, S. and Choi, J. (2022) 'Twitter Attribute Classification With Q-Learning on Bitcoin Price Prediction', *IEEE Access*, Vol. 10, pp. 96136–96148, doi: 10.1109/ACCESS.2022.3205129.
- Parekh, R., Patel, N.P., Thakkar, N., Gupta, R., Tanwar, S., Sharma, G., Davidson, I.E. and Sharma, R. (2022) 'DL-GuesS: Deep Learning and Sentiment Analysis-Based Cryptocurrency Price Prediction', *IEEE Access*, Vol. 10, pp. 35398–35409, doi: 10.1109/ACCESS.2022.3163305.
- Politis, D.N. and Romano, J.P. (1994) 'The Stationary Bootstrap', *Journal of the American Statistical Association*, Vol. 89, No. 428, pp. 1303–1313, doi: 10.1080/01621459.1994.10476870.
- Sami, H.M., Shah Jalal, M., Faisal Kabir, M. and Nurul Huda, M. (2025) 'IntPort: An Intelligent Portfolio Construction Technique Based on Financial Forecasting by Statistical Average Method', *IEEE Access*, Vol. 13, pp. 35355–35375, doi: 10.1109/ACCESS.2025.3543741.

- Shang, L. (2025) 'Sentiment-Driven Bitcoin Price Range Forecasting: Enhancing CART Decision Trees With High-Dimensional Indicators and Twitter Dynamics', *IEEE Access*, Vol. 13, pp. 60508–60518, doi: 10.1109/ACCESS.2025.3557186.
- Tai, W., Zhong, T., Mo, Y. and Zhou, F. (2022) 'Learning Sentimental and Financial Signals With Normalizing Flows for Stock Movement Prediction', *IEEE Signal Processing Letters*, Vol. 29, pp. 414–418, doi: 10.1109/LSP.2021.3135793.
- Vargas, G., Silvestre, L., Júnior, L.R. and Rocha, H. (2022) 'B3 Stock Price Prediction Using LSTM Neural Networks and Sentiment Analysis', *IEEE Latin America Transactions*, Vol. 20, No. 7, pp. 1067–1074, doi: 10.1109/TLA.2021.9827469.
- Wang, C., Xiao, Z., Zhao, D.-S., Ni, D. and Li, L. (2019) 'Research on Economic Recession Prediction Model From the Multiple Behavioral Features Perspective', *IEEE Access*, Vol. 7, pp. 83363–83371, doi: 10.1109/ACCESS.2019.2924267.
- Wang, F., Li, M., Mei, Y. and Li, W. (2020) 'Time Series Data Mining: A Case Study With Big Data Analytics Approach', *IEEE Access*, Vol. 8, pp. 14322–14328, doi: 10.1109/ACCESS.2020.2966553.
- Wang, J. and Chen, Z. (2024) 'SPCM: A Machine Learning Approach for Sentiment-Based Stock Recommendation System', *IEEE Access*, Vol. 12, pp. 14116–14129, doi: 10.1109/ACCESS.2024.3357114.
- Yang, Y. and Yang, Y. (2020) 'Hybrid Method for Short-Term Time Series Forecasting Based on EEMD', *IEEE Access*, Vol. 8, pp. 61915–61928, doi: 10.1109/ACCESS.2020.2983588.
- Zhang, D., Chen, S., Liwen, L. and Xia, Q. (2020) 'Forecasting Agricultural Commodity Prices Using Model Selection Framework With Time Series Features and Forecast Horizons', *IEEE Access*, Vol. 8, pp. 28197–28209, doi: 10.1109/ACCESS.2020.2971591.
- Zhang, C., Sjarif, N.N.A. and Ibrahim, R.B. (2022) 'Decision Fusion for Stock Market Prediction: A Systematic Review', *IEEE Access*, Vol. 10, pp. 81364–81379, doi: 10.1109/ACCESS.2022.3195942.
- Zhang, Z., Choo, W.C. and Arasan, J. (2025) 'Does the optimal model always perform the best? A combined approach for interval forecasting', *International Journal of Applied Decision Sciences*, Vol. 18, No. 1, pp. 64–83, doi: 10.1504/IJADS.2025.143079.